GLOBAL SHADOW ESTIMATES AND HAUSDORFF MEASURE CRITERIA IN HIGHER RANK

DONGRYUL M. KIM AND HEE OH

Abstract. Patterson–Sullivan measures encode the distribution of orbits of discrete group actions near the boundary. We prove a global shadow lemma for Patterson–Sullivan measures associated to relatively Morse subgroups of higher-rank semisimple Lie groups. The estimate is uniform for shadows centered at arbitrary points in a Gromov model, including points deep in the cuspidal part, and expresses the contribution of each cusp in terms of peripheral growth. This extends the global shadow lemma of Stratmann–Velani for geometrically finite real hyperbolic groups. As applications, we establish local doubling and reverse doubling. Under Zariski density, we give necessary and sufficient criteria for Patterson–Sullivan measures to agree with the associated Hausdorff measures and to be Ahlfors regular. For symmetric linear forms, these criteria hold without Zariski density. We verify the criteria explicitly for cusped Hitchin representations and real-hyperbolic self-joinings.

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1. Introduction

Patterson–Sullivan measures relate orbit growth to the geometry of limit sets. In the presence of cusps, their local behavior also reflects the growth of peripheral subgroups. A global shadow estimate describes this dependence uniformly, including for shadows centered arbitrarily deep in cusp horoballs. In this paper we establish such an estimate for relatively Morse subgroups of

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higher-rank semisimple Lie groups. We use it to give criteria characterizing when these measures agree, up to scale, with the associated Hausdorff measures and when they are Ahlfors regular. We give explicit applications to cusped Hitchin representations and real-hyperbolic self-joinings. We begin with the classical real hyperbolic case.

Let $\Gamma < \mathrm{Isom}^+(\mathbb{H}^n)$ be a non-elementary discrete subgroup, let $o \in \mathbb{H}^n$, and let $\Lambda \subset \partial\mathbb{H}^n$ be the limit set of Γ. Patterson [21] and Sullivan [25] constructed a Borel probability measure ν supported on Λ whose transformation rule is governed by the critical exponent $\delta_\Gamma > 0$:

$$\frac{d\gamma_*\nu}{d\nu}(\xi) = e^{\delta_\Gamma \beta_\xi(o,\gamma o)} \quad \text{for all } \gamma \in \Gamma \text{ and } \nu\text{-a.e. } \xi \in \Lambda.$$

Here β_ξ denotes the Busemann function. The measure ν is now referred to as the Patterson–Sullivan measure of Γ.

A fundamental feature of this measure is Sullivan's shadow lemma. For $x, y \in \mathbb{H}^n$ and $R > 0$, let $O_R(x, y) \subset \partial\mathbb{H}^n$ denote the shadow of the ball $B(y, R)$ seen from x, namely the set of endpoints ξ such that the geodesic ray $[x, \xi] \subset \mathbb{H}^n$ intersects the R-neighborhood of y.

Theorem 1.1 (Shadow Lemma [25, Proposition 3])**.** *For all sufficiently large $R > 0$, there exists $C > 1$ such that*

$$C^{-1} e^{-\delta_\Gamma d(o,\gamma o)} \le \nu(O_R(o, \gamma o)) \le C e^{-\delta_\Gamma d(o,\gamma o)} \qquad (\gamma \in \Gamma).$$

Thus the Patterson–Sullivan measure of a shadow is comparable to the exponential of minus the orbit distance: $\nu(O_R(o, \gamma o)) \asymp e^{-\delta_\Gamma d(o,\gamma o)}$, with implied constants independent of γ. This estimate is the basic prototype for the global shadow lemma proved in this paper.

Geometrically finite groups. For geometrically finite groups, Sullivan's shadow lemma admits a global version due to Stratmann–Velani [24]. Let $\Gamma < \mathrm{Isom}^+(\mathbb{H}^n)$ be geometrically finite with limit set $\Lambda \subset \partial\mathbb{H}^n$, and let $\mathrm{hull}\,\Lambda \subset \mathbb{H}^n$ be its convex hull. Choose a Γ-invariant family $\mathcal{B}$ of pairwise disjoint cusp horoballs in $\mathrm{hull}\,\Lambda$ whose complement has compact quotient. Let $\mathcal{P}$ be a set of representatives of the corresponding maximal parabolic subgroups, and write δ_{P} for the critical exponent of $\mathsf{P} \in \mathcal{P}$. It is known that $\delta_{\mathsf{P}} < \delta_\Gamma$ ([25], [26]).

Theorem 1.2 (Global Shadow Lemma [24, Theorem 2])**.** *For all sufficiently large $R > 0$, the following holds. Let $\xi \in \Lambda$ and $x \in [o, \xi]$. If x lies in a horoball $B \in \mathcal{B}$ whose stabilizer is conjugate to $\mathsf{P} \in \mathcal{P}$, then*

$$\nu(O_R(o, x)) \asymp e^{-\delta_\Gamma d(o,x)} e^{(2\delta_{\mathsf{P}} - \delta_\Gamma) d(\Gamma o, x)},$$

with implied constants independent of ξ, x, and B.

In $\mathrm{hull}\,\Lambda$ but away from the cusp horoballs, the quantity $d(\Gamma o, x)$ is uniformly bounded, and the estimate reduces to the usual shadow lemma $\nu(O_R(o, x)) \asymp e^{-\delta_\Gamma d(o,x)}$.

Relatively Anosov and relatively Morse groups. Relatively Anosov subgroups provide a higher-rank analogue of geometrically finite Kleinian groups. Let G be a connected semisimple real algebraic group with Cartan decomposition $G = KA^+K$ where K is a maximal compact subgroup. Let $X := G/K$ denote the associated Riemannian symmetric space. Let $\mathfrak{a}^+ := \log A^+$, let Π be the corresponding set of simple roots, and fix a non-empty subset $\theta \subset \Pi$. We denote by P_θ the associated standard parabolic subgroup and by

$$\mathcal{F}_\theta := G/P_\theta$$

the corresponding flag variety. We also set

$$\mathfrak{a}_\theta := \bigcap_{\alpha \in \Pi - \theta} \ker \alpha$$

and regard $\mathfrak{a}_\theta^*$ as a subspace of $\mathfrak{a}^*$ via the canonical projection $p_\theta : \mathfrak{a} \to \mathfrak{a}_\theta$. After replacing θ by $\theta \cup \mathrm{i}(\theta)$, if necessary, we assume that

$$\theta = \mathrm{i}(\theta),$$

where i is the opposition involution on $\mathfrak{a}$.

Let $\Gamma < G$ be a discrete subgroup which is hyperbolic relative to a finite collection $\mathcal{P}$ of finitely generated infinite subgroups. We write $\partial(\Gamma, \mathcal{P})$ for its Bowditch boundary. A Gromov model for $(\Gamma, \mathcal{P})$ is a proper geodesic Gromov hyperbolic space Y on which Γ acts properly discontinuously, together with a Γ-invariant collection $\mathcal{B}$ of disjoint horoballs whose stabilizers are conjugates of subgroups in $\mathcal{P}$, and on whose complement Γ acts cocompactly. We identify ∂Y with $\partial(\Gamma, \mathcal{P})$, and we assume that it contains at least three points, i.e., $(\Gamma, \mathcal{P})$ is non-elementary.

We say that Γ is θ-Anosov relative to $\mathcal{P}$ if it is θ-regular and admits a transverse Γ-equivariant boundary map

$$\zeta : \partial Y \to \mathcal{F}_\theta.$$

See section 4 for its precise definition. The image of this map is the θ-limit set, denoted by Λ_θ.

For $\psi \in \mathfrak{a}_\theta^*$, a (Γ, ψ)-Patterson–Sullivan measure is a Borel probability measure ν on Λ_θ satisfying

$$\frac{d\gamma_*\nu}{d\nu}(\xi) = e^{\psi(\beta^\theta_\xi(e,\gamma))} \quad \text{for all } \gamma \in \Gamma \text{ and } \nu\text{-a.e. } \xi \in \Lambda_\theta,$$

where β^θ is the $\mathfrak{a}_\theta$-valued Busemann map; see (2.3). This notion of higher-rank Patterson–Sullivan measure was introduced by Quint [22].

Let $\mu : G \to \mathfrak{a}^+$ be the Cartan projection. Let $\mathcal{L}_\Gamma \subset \mathfrak{a}^+$ denote the asymptotic cone of the Cartan projection $\mu(\Gamma)$ of Γ, called the limit cone of Γ. The shadow lemma for Patterson–Sullivan measures was proved for shadows in the higher-rank symmetric space $X = G/K$ in [19, Lemma 7.8] and [18, Lemma 7.2]. For relatively θ-Anosov subgroups, the compatibility between shadows in the Gromov model Y and shadows in X was established

in [17, Proposition 5.7].[1] Combining these results gives the following orbit-shadow estimate.

Theorem 1.3 (Shadow Lemma)**.** *Let $\Gamma < G$ be θ-Anosov relative to $\mathcal{P}$, and fix $o_Y \in Y$. For $\psi \in \mathfrak{a}_\theta^*$, let ν be a (Γ, ψ)-Patterson–Sullivan measure on Λ_θ. Then, for all sufficiently large $R > 0$,*

$$\nu\big(\zeta(O_R(o_Y, \gamma o_Y))\big) \asymp e^{-\psi(\mu(\gamma))} \qquad (\gamma \in \Gamma),$$

with implied constants independent of γ.

The relatively Morse condition strengthens the relative Anosov condition by requiring the relative geometry of Y to be realized inside the symmetric space. We say that Γ is θ-Morse relative to $\mathcal{P}$ if there exists a Γ-equivariant quasi-isometric embedding

$$f : Y \to X$$

such that, writing $\mathcal{L}_f \subset \mathfrak{a}^+$ for the asymptotic cone of the Cartan projections

$$\{\mu(f(x)^{-1} f(y)) : x, y \in Y\},$$

we have

$$\mathcal{L}_f \cap \ker \alpha = \{0\} \quad \text{for every } \alpha \in \theta.$$

Thus being relative Anosov gives a boundary map into the flag variety, whereas the relatively Morse condition gives a coarse geometric model in the symmetric space whose Cartan projections stay uniformly away from the walls indexed by θ. By the higher-rank Morse lemma of Kapovich-Leeb-Porti [15, Theorem 1.4], the map f extends continuously to a transverse Γ-equivariant homeomorphism

$$f : \partial Y \to \Lambda_\theta.$$

Fix a basepoint $o_Y \in Y$. For $\psi \in \mathfrak{a}_\theta^*$ positive on $\mathcal{L}_f - \{0\}$, define

$$\mathsf{d}_\psi(x, y) := \psi(\mu(f(x)^{-1} f(y))) \quad \text{for } x, y \in Y,$$

and, for subsets $E, F \subset Y$, set

$$\mathsf{d}_\psi(E, F) := \inf_{x \in E,\, y \in F} \mathsf{d}_\psi(x, y).$$

Theorem 1.3 estimates shadows centered at orbit points. The main result of this paper is a global version for relatively Morse groups, where the center of the shadow may be an arbitrary point along a geodesic ray in the Gromov model.

For a subgroup $H < \Gamma$, denote by $\delta_\psi(H)$ the critical exponent of the Poincaré series

$$s \mapsto \sum_{\gamma \in H} e^{-s\psi(\mu(\gamma))}.$$

In the setting below, the strict inequality $\delta_\psi(\mathsf{P}) < \delta_\psi(\Gamma)$ was proved by Canary–Zhang–Zimmer [9].

[1] The reference [17] treats a specific Gromov model, but the same proof applies to the Gromov models considered here.

The following theorem is the higher-rank relatively Morse analogue of the global shadow lemma of Stratmann–Velani [24].

Theorem 1.4 (Global Shadow Lemma)**.** *Let $\Gamma < G$ be θ-Morse relative to $\mathcal{P}$, and let $\psi \in \mathfrak{a}_\theta^*$ be such that $\psi > 0$ on $\mathcal{L}_f - \{0\}$ and $\delta_\psi(\Gamma) = 1$. Let ν be a (Γ, ψ)-Patterson–Sullivan measure on Λ_θ. Then there exists $C_0 > 0$ such that, for all sufficiently large $R > 0$, the following holds.*

Let $\xi \in \partial Y$ and $x \in [o_Y, \xi]$. Suppose that $x \in B$ for some horoball $B \in \mathcal{B}$ whose stabilizer is conjugate to $\mathsf{P} \in \mathcal{P}$. Then

$$\nu(f(O_R(o_Y, x))) \asymp e^{-\mathsf{d}_\psi(o_Y, x)} e^{\mathsf{d}_\psi(\Gamma o_Y, x)} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\Gamma o_Y, x)} \cdot \big(C_0 + \mathsf{d}_{\bar\psi}(\Gamma o_Y, x)\big)^{a_{\bar\psi}(\mathsf{P})},$$

where $\bar\psi = \frac{\psi + \psi \circ \mathsf{i}}{2}$ and $a_{\bar\psi}(\mathsf{P})$ is a non-negative integer depending on $\bar\psi$ and P. The implied constants are independent of ξ, x, and B.

See Remark 8.2 for estimates in terms of the visual quasi-metric on Λ_θ.

Here $a_{\bar\psi}(\mathsf{P})$ is the exponent of the polynomial factor in the growth of P measured by $\bar\psi$. It vanishes when P is virtually cyclic (Theorem 5.1). It would be interesting to know whether the relatively Morse hypothesis forces $a_{\bar\psi}(\mathsf{P}) = 0$ for every $\mathsf{P} \in \mathcal{P}$.

If x lies in the thick part $Y - \bigcup_{B \in \mathcal{B}} B$, then Theorem 1.4 recovers the usual orbit-shadow estimate. The theorem therefore identifies the precise additional factor that appears when x enters a cusp.

Remark 1.5. Bray–Tiozzo [6, Theorem 1.4] proved a global shadow lemma for geometrically finite actions on proper hyperbolic metric spaces under a tempered parabolic growth assumption. Their measures are quasi-conformal with respect to the Busemann functions of the hyperbolic space. In our setting, although shadows are taken in the Gromov model, the Busemann maps, Patterson–Sullivan measures, and critical exponents come from the ambient higher-rank Lie group.

Example 1.6. The relatively Morse hypothesis holds for the following families.

(1) Let $G = \prod_{i=1}^k \mathrm{SO}^\circ(n_i, 1)$, $n_i \geq 2$. It follows from the work of Bowditch [3] and Yaman [28] that a discrete subgroup $\Gamma < G$ is relatively Π-Anosov if and only if there exists a geometrically finite subgroup $\Gamma_1 < \mathrm{SO}^\circ(n_1, 1)$ and geometrically finite type-preserving representations $\rho_i : \Gamma_1 \to \mathrm{SO}^\circ(n_i, 1)$, $2 \leq i \leq k$, so that the diagonal embedding $(\mathrm{id} \times \rho_2 \times \cdots \times \rho_k)(\Gamma_1)$ is a finite-index subgroup of Γ. The work of Tukia [27] (also see [10]) then implies that Γ is relatively Π-Morse. Moreover, there exists a Morse embedding f such that $\mathcal{L}_f = \mathcal{L}_\Gamma$.
(2) We consider a relatively θ-Anosov $\Gamma < \mathrm{SL}(n, \mathbb{R})$, $n \geq 2$, with peripheral subgroups $\mathcal{P}$, such that $(\Gamma, \mathcal{P})$ is isomorphic to a geometrically

finite Fuchsian group. In this case, Zhu–Zimmer showed that Γ is relatively θ-Morse ([30, Corollary 1.14], [29, Proposition 11.3]). They explicitly constructed a Morse embedding f such that $\mathcal{L}_f = \mathcal{L}_\Gamma$ in [29, Sections 6, 10, 11].

Local properties of Patterson–Sullivan measures. The global shadow lemma also gives local information on Patterson–Sullivan measures. For distinct $\xi, \eta \in \Lambda_\theta$, set

$$d_\psi(\xi, \eta) := e^{-\psi(\mathcal{G}^\theta(\xi,\eta))}$$

where $\mathcal{G}^\theta$ is the $\mathfrak{a}_\theta$-valued Gromov product. This is a higher-rank visual quasi-metric on Λ_θ; denote by $B_\psi(\xi, r)$ the corresponding balls. As applications, we obtain the following uniform local estimates for Patterson–Sullivan measures.

Corollary 1.7 (Uniform local estimates)**.** *Let Γ, ψ, ν be as in Theorem 1.4. For every $\kappa \geq 1$, there exists $L_\kappa > 1$ and $C_\kappa > 1$ such that*

$$C_\kappa^{-1}\nu(B_\psi(\xi, L_\kappa r)) \leq \nu(B_\psi(\xi, r)) \leq \kappa^{-1}\nu(B_\psi(\xi, L_\kappa r))$$

for all $\xi \in \Lambda_\theta$ and all $0 < r \leq L_\kappa^{-1}$.

This corollary follows immediately from Theorem 8.3.

Finally, the global estimate gives a sharp criterion for the Patterson–Sullivan measure to be a Hausdorff measure. Denote by $\mathcal{H}_\psi$ the one-dimensional Hausdorff measure associated to d_ψ. For Anosov groups, Dey–Kim–Oh [12, Theorem 1.1] proved that Patterson–Sullivan measures associated to symmetric linear forms are Ahlfors 1-regular and agree, up to scale, with the associated Hausdorff measures. Our results recover their theorem when $\mathcal{P} = \emptyset$ and determine precisely how peripheral growth affects these conclusions in the presence of cusps. In particular, identification with Hausdorff measure can persist even when Ahlfors regularity fails.

Theorem 1.8 (Hausdorff measure)**.** *Let Γ, ψ, ν be as in Theorem 1.4. Then $\mathcal{H}_\psi(\Lambda_\theta) < \infty$. If Γ is Zariski dense, then ν is a positive multiple of $\mathcal{H}_\psi$ if and only if $\psi = \psi \circ \mathrm{i}$ and, for every $\mathsf{P} \in \mathcal{P}$, either*

$$\delta_\psi(\mathsf{P}) < \frac{1}{2}, \quad \text{or} \quad \delta_\psi(\mathsf{P}) = \frac{1}{2} \text{ and } a_\psi(\mathsf{P}) = 0.$$

For symmetric ψ, this peripheral criterion is also necessary and sufficient without assuming that Γ is Zariski dense.

The theorem is proved in Section 9. For virtually cyclic peripheral subgroups, the peripheral condition reduces to $\delta_\psi(\mathsf{P}) \leq 1/2$ for every P. If Γ is Zariski dense, we also show that ν is Ahlfors regular if and only if $\psi = \psi \circ \mathrm{i}$ and

$$\delta_\psi(\mathsf{P}) = \frac{1}{2} \quad \text{and} \quad a_\psi(\mathsf{P}) = 0 \qquad \text{for every } \mathsf{P} \in \mathcal{P};$$

in that case its Ahlfors exponent is 1 (Corollary 9.5). For symmetric ψ, this Ahlfors regularity criterion also holds without Zariski density. Thus, in the

symmetric case, peripheral growth determines both whether the Hausdorff measure is positive and whether its local mass is uniformly comparable to the radius.

The hypotheses can be checked explicitly in familiar families. For a cusped Hitchin representation of a torsion-free nonuniform lattice in $\mathrm{PSL}(2,\mathbb{R})$ into $\mathrm{PSL}(2m,\mathbb{R})$, with $m \geq 2$, the middle simple root $\psi = \alpha_m$ is symmetric and satisfies

$$\delta_\psi(\Gamma) = 1, \qquad \delta_\psi(\mathsf{P}) = \frac{1}{2}, \qquad a_\psi(\mathsf{P}) = 0$$

for every peripheral subgroup P. Consequently, ν_ψ is Ahlfors 1-regular (Corollary 9.6). For self-joinings in products of real hyperbolic groups, the peripheral exponents can also be computed directly from the cusp ranks and the coefficients of ψ (Corollary 9.7). In particular, self-joinings of two finite-area Fuchsian representations that are not conjugate in $\mathrm{PGL}(2,\mathbb{R})$ give examples in which ν_ψ is locally doubling and locally reverse doubling but $\mathcal{H}_\psi(\Lambda_\theta) = 0$ (Example 9.8).

AI disclosure: The argument establishing the necessity of the condition in Theorem 9.1 was suggested by an AI tool.

2. Preliminaries

Let G be a connected semisimple real algebraic group. Let P be a minimal parabolic subgroup with a fixed Langlands decomposition $P = MAN$ where A is a maximal real split torus of G, M is the maximal compact subgroup of P commuting with A and N is the unipotent radical of P. Let $\mathfrak{g}$ and $\mathfrak{a}$ denote, respectively, the Lie algebras of G and A. Fix a positive Weyl chamber $\mathfrak{a}^+ \subset \mathfrak{a}$ so that $\log N$ consists of positive root subspaces and set $A^+ = \exp \mathfrak{a}^+$. We fix a maximal compact subgroup $K < G$ such that the Cartan decomposition $G = KA^+K$ holds. We denote by $\mu : G \to \mathfrak{a}^+$ the Cartan projection defined by the condition $g \in K \exp \mu(g) K$ for $g \in G$. Let $X = G/K$ be the associated Riemannian symmetric space and $o = [K] \in X$. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{g}$ with associated norm $\|\cdot\|$. This induces the left G-invariant Riemannian metric d on X.

Lemma 2.1. [1, Lemma 4.6] *For any compact subset $Q \subset G$, there exists $C = C(Q) > 0$ such that for all $g \in G$,*

$$\sup_{q_1, q_2 \in Q} \|\mu(q_1 g q_2) - \mu(g)\| \leq C.$$

Let Π denote the set of simple roots determined by $\mathfrak{a}^+$. We fix a nonempty subset $\theta \subset \Pi$. Let P_θ denote the corresponding standard parabolic subgroup with the convention that $P_\Pi = P$, and set $\mathcal{F}_\theta := G/P_\theta$. We also set

$$\mathfrak{a}_\theta := \bigcap_{\alpha \in \Pi - \theta} \ker \alpha \quad \text{and} \quad \mathfrak{a}_\theta^+ := \mathfrak{a}_\theta \cap \mathfrak{a}^+.$$

Let

$$p_\theta : \mathfrak{a} \to \mathfrak{a}_\theta \tag{2.1}$$

denote the projection invariant under all Weyl elements fixing $\mathfrak{a}_\theta$ pointwise. We write $\mu_\theta := p_\theta \circ \mu : G \to \mathfrak{a}_\theta^+$. We identify $\mathfrak{a}_\theta^* = \mathrm{Hom}(\mathfrak{a}_\theta, \mathbb{R})$ with the subspace of $\mathfrak{a}^*$ consisting of linear forms invariant under p_θ.

Abusing notation, for $p, q \in X$, we set

$$\mu(p) := \mu(g) \quad \text{and} \quad \mu(p^{-1}q) := \mu(g^{-1}h)$$

for $g, h \in G$ such that $go = p$ and $ho = q$. This definition is independent of the choice of g and h, and we similarly define $\mu_\theta(p)$ and $\mu_\theta(p^{-1}q)$.

Let $w_0 \in K$ represent the longest Weyl element. The opposition involution is

$$\mathrm{i} := -\mathrm{Ad}_{w_0} : \mathfrak{a} \to \mathfrak{a},$$

and it induces an involution of Π, again denoted by i.

Limit set. The subgroup K acts transitively on $\mathcal{F}_\theta$, and hence $\mathcal{F}_\theta \simeq K/M_\theta$ where $M_\theta := P_\theta \cap K$. Set $\xi_\theta := [M_\theta] \in \mathcal{F}_\theta$.

Definition 2.2. For a sequence $g_i \in G$ and $\xi \in \mathcal{F}_\theta$, we write $\lim_{i\to\infty} g_i = \xi$ and say g_i *converges* to ξ if

- for each $\alpha \in \theta$, $\alpha(\mu(g_i)) \to \infty$ as $i \to \infty$;
- $\lim_{i\to\infty} \kappa_i \xi_\theta = \xi$ in $\mathcal{F}_\theta$ for some $\kappa_i \in K$ such that $g_i \in \kappa_i A^+ K$.

The *θ-limit set* of a discrete subgroup Γ can be defined as follows:

$$\Lambda_\theta = \Lambda_\theta(\Gamma) := \{\lim \gamma_i \in \mathcal{F}_\theta : \gamma_i \in \Gamma\} \tag{2.2}$$

where $\lim \gamma_i$ is defined as in Definition 2.2. If Γ is Zariski dense, this is the unique Γ-minimal subset of $\mathcal{F}_\theta$ [1].

Busemann map. The *$\mathfrak{a}$-valued Busemann map* $\beta : \mathcal{F}_\Pi \times G \times G \to \mathfrak{a}$ is defined as follows: for $\xi \in \mathcal{F}_\Pi$ and $g, h \in G$,

$$\beta_\xi(g, h) := \sigma(g^{-1}, \xi) - \sigma(h^{-1}, \xi)$$

where $\sigma(g^{-1}, \xi) \in \mathfrak{a}$ is the unique element such that $g^{-1}k \in K \exp(\sigma(g^{-1}, \xi))N$ for any $k \in K$ with $\xi = kP$. For $(\xi, g, h) \in \mathcal{F}_\theta \times G \times G$, we define

$$\beta_\xi^\theta(g, h) := p_\theta(\beta_{\xi_0}(g, h)) \tag{2.3}$$

for any $\xi_0 \in \mathcal{F}_\Pi$ projecting to ξ. This is well-defined independent of the choice of ξ_0 [22, Lemma 6.1]. Moreover, since product map $K \times A \times N \to G$ is a diffeomorphism, the Busemann maps are continuous.

Gromov product. Two points $\xi \in \mathcal{F}_\theta$ and $\eta \in \mathcal{F}_{\mathrm{i}(\theta)}$ are said to be *transverse*, or *in general position* if

$$\xi = gP_\theta \quad \text{and} \quad \eta = gw_0P_{\mathrm{i}(\theta)} \quad \text{for some } g \in G. \tag{2.4}$$

We set

$$\mathcal{F}_\theta^{(2)} = \{(\xi,\eta) \in \mathcal{F}_\theta \times \mathcal{F}_{\mathrm{i}(\theta)} : \xi,\eta \text{ are in general position}\} \tag{2.5}$$

which is the unique open G-orbit in $\mathcal{F}_\theta \times \mathcal{F}_{\mathrm{i}(\theta)}$ under the diagonal G-action.

For $(\xi,\eta) \in \mathcal{F}_\theta^{(2)}$, we define the $\mathfrak{a}_\theta$*-valued Gromov product* as

$$\mathcal{G}^\theta(\xi,\eta) = \frac{1}{2}\left(\beta_\xi^\theta(e,g) + \mathrm{i}\,\beta_\eta^{\mathrm{i}(\theta)}(e,g)\right) \tag{2.6}$$

where $g \in G$ satisfies $(gP_\theta, gw_0P_{\mathrm{i}(\theta)}) = (\xi,\eta)$. This is independent of the choice of g [18, Lemma 9.13].

Patterson–Sullivan measures. For $\psi \in \mathfrak{a}_\theta^*$, a (Γ,ψ)-*conformal measure* is a Borel probability measure on $\mathcal{F}_\theta$ such that

$$\frac{d\gamma_*\nu}{d\nu}(\xi) = e^{\psi(\beta_\xi^\theta(e,\gamma))} \quad \text{for all } \gamma \in \Gamma \text{ and } \nu\text{-a.e.}\xi \in \mathcal{F}_\theta \tag{2.7}$$

where $\gamma_*\nu(D) = \nu(\gamma^{-1}D)$ for any Borel subset $D \subset \mathcal{F}_\theta$ and β_ξ^θ denotes the $\mathfrak{a}_\theta$-valued Busemann map defined in (2.3). A (Γ,ψ)-conformal measure supported on Λ_θ is called a (Γ,ψ)-*Patterson–Sullivan measure*.

3. Relatively hyperbolic groups and Gromov models

In this section, we recall relatively hyperbolic groups, Gromov hyperbolic spaces, and Gromov models.

Relatively hyperbolic groups. Let Γ be a countable group acting by homeomorphisms on a compact metrizable space $\mathcal{X}$. The action is called a *convergence group action* if, for every sequence of distinct elements $\gamma_n \in \Gamma$, there exist a subsequence γ_{n_k} and points $a,b \in \mathcal{X}$ such that $\gamma_{n_k}(x)$ converges to a for all $x \in \mathcal{X} - \{b\}$, uniformly on compact subsets.

An infinite-order element $\gamma \in \Gamma$ is called *loxodromic* if it fixes exactly two points of $\mathcal{X}$, and *parabolic* if it fixes exactly one point. An infinite subgroup $\mathsf{P} < \Gamma$ is called *parabolic* if it fixes a point of $\mathcal{X}$ and every infinite-order element of P is parabolic.

A point $\xi \in \mathcal{X}$ is called a *conical limit point* if there exist a sequence of distinct elements $\gamma_n \in \Gamma$ and distinct points $a,b \in \mathcal{X}$ such that

$$\gamma_n^{-1}\xi \to a \quad \text{and} \quad \gamma_n^{-1}\eta \to b \quad \text{for all } \eta \in \mathcal{X} - \{\xi\}.$$

A point $\xi \in \mathcal{X}$ is called a *parabolic limit point* if it is fixed by a parabolic subgroup of Γ. Such a point is called *bounded parabolic* if

$$\mathrm{Stab}_\Gamma(\xi)\backslash(\mathcal{X} - \{\xi\})$$

is compact. The action of Γ on $\mathcal{X}$ is called a *geometrically finite convergence group action* if every point of $\mathcal{X}$ is either conical or bounded parabolic. A

typical example is the action of a geometrically finite Kleinian group on its limit set.

Let Γ be a finitely generated group and let $\mathcal{P}$ be a finite collection of finitely generated infinite subgroups of Γ. We say that Γ is *hyperbolic relative to* $\mathcal{P}$, or that $(\Gamma, \mathcal{P})$ is *relatively hyperbolic*, if Γ admits a geometrically finite convergence group action on a compact perfect metrizable space $\mathcal{X}$ whose maximal parabolic subgroups are precisely

$$\mathcal{P}^\Gamma := \{\gamma \mathsf{P} \gamma^{-1} : \mathsf{P} \in \mathcal{P},\ \gamma \in \Gamma\}.$$

Bowditch [5] showed that, if Γ is hyperbolic relative to $\mathcal{P}$, then the space $\mathcal{X}$ satisfying the above condition is unique up to Γ-equivariant homeomorphism. This space is called the *Bowditch boundary* and is denoted by $\partial(\Gamma, \mathcal{P})$. Since $\partial(\Gamma, \mathcal{P})$ is assumed to be perfect, we have $\#\partial(\Gamma, \mathcal{P}) \geq 3$; equivalently, $(\Gamma, \mathcal{P})$ is non-elementary.

Gromov hyperbolic spaces. A proper geodesic metric space (Y, d) is called *Gromov hyperbolic* if there exists $\delta > 0$ such that every geodesic triangle in Y is δ-thin; that is, each side is contained in the δ-neighborhood of the union of the other two sides. The Gromov boundary ∂Y is the set of equivalence classes of geodesic rays, where two rays are equivalent if they have finite Hausdorff distance. We write

$$\overline{Y} := Y \cup \partial Y$$

for the corresponding compactification.

For $C_1, C_2 \geq 1$ and an interval $I \subset \mathbb{R}$, a map $\sigma : I \to Y$ is called a (C_1, C_2)-quasi-geodesic if

$$C_1^{-1}|t - s| - C_2 \leq d(\sigma(t), \sigma(s)) \leq C_1|t - s| + C_2 \quad \text{for all } t, s \in I.$$

We also call the image $\sigma(I)$ a (C_1, C_2)-quasi-geodesic. We shall use the following standard stability property.

Lemma 3.1. *For any $C_1, C_2 \geq 1$, there exists $R > 0$ such that any two (C_1, C_2)-quasi-geodesics in Y with the same endpoints in $\overline{Y}$ have Hausdorff distance at most R.*

We use the following notation.

Definition 3.2. Let $y, y_1, y_2 \in Y$, let $z_1, z_2 \in \overline{Y}$, and let $R > 0$.

(1) Let
$$B(y, R) = \{x \in Y : d(y, x) < R\}$$
denote the ball of radius R centered at y.

(2) The notation $[y_1, y_2]$ denotes a choice of geodesic in Y connecting y_1 to y_2. Such a geodesic need not be unique.

(3) The shadow $O_R(y_1, y_2) \subset \partial Y$ is

$$O_R(y_1, y_2) = \{\xi \in \partial Y :\ \text{some geodesic ray } [y_1, \xi] \text{ intersects } B(y_2, R)\}.$$

(4) We denote by $\pi_{z_1, z_2}(y)$ the set of nearest-point projections of y to all geodesics between z_1 and z_2.

(5) We denote by $\pi_{[z_1,z_2]}(y)$ the set of nearest-point projections of y to the chosen geodesic $[z_1, z_2]$.

It is well known that, for any $x \in \pi_{z_1,z_2}(y)$, the concatenation $[y,x]\cup[x,z_1]$ is a $(1, O(\delta))$-quasi-geodesic, where $O(\delta)$ denotes a constant depending only on δ.

Given $o_Y \in Y$ and two distinct points $\xi, \eta \in Y \cup \partial Y$, define their Gromov product with respect to o_Y by

$$\langle \xi, \eta \rangle_{o_Y} := \sup \liminf_{i,j\to\infty} \frac{1}{2}\big(d(o_Y, x_i) + d(o_Y, y_j) - d(x_i, y_j)\big),$$

where the supremum is taken over all sequences $x_i, y_j \in Y$ such that $x_i \to \xi$ and $y_j \to \eta$. This quantity measures the distance from o_Y to $\pi_{\xi,\eta}(o_Y)$ up to a uniform additive error depending only on δ. We shall use the standard inequality

$$\langle \xi, \eta \rangle_{o_Y} \geq \min\left(\langle \xi, \zeta \rangle_{o_Y}, \langle \zeta, \eta \rangle_{o_Y}\right) - O(\delta) \tag{3.1}$$

for all $\xi, \eta, \zeta \in Y \cup \partial Y$.

The following standard comparison between shadows and Gromov products will be used repeatedly.

Lemma 3.3. *Let $o_Y \in Y$, $\xi \in \partial Y$, and $\xi_t \in [o_Y, \xi]$ be such that $d(o_Y, \xi_t) = t$ for $t \geq 0$. For any $R > 0$:*

(1) *if $\eta \in O_R(o_Y, \xi_t)$, then*

$$\langle \xi, \eta \rangle_{o_Y} \geq t - R - O(\delta);$$

(2) *if $\langle \xi, \eta \rangle_{o_Y} \geq t - R + O(\delta)$, then*

$$\eta \in O_R(o_Y, \xi_t).$$

Proof. We first prove (1). Let $\eta \in O_R(o_Y, \xi_t)$ and let $y \in \pi_{[\xi,\eta]}(o_Y)$. It suffices to consider the case $\langle \xi, \eta \rangle_{o_Y} \leq t$. In this case, ξ_t is $O(\delta)$-close to $[\xi, \eta]$. Choose $x \in [\xi, \eta]$ with $d(\xi_t, x) < O(\delta)$. Since $\eta \in O_R(o_Y, \xi_t)$, Lemma 3.1 implies that ξ_t is $R + O(\delta)$-close to $[o_Y, y] \cup [y, \eta]$. Hence x is also $R + O(\delta)$-close to $[o_Y, y] \cup [y, \eta]$.

If x is $R + O(\delta)$-close to $[y, \eta]$, then $d(x, y) < R + O(\delta)$, since both x and $[y, \eta]$ lie on the geodesic $[\xi, \eta]$. Therefore

$$d(o_Y, y) \geq d(o_Y, x) - R - O(\delta) \geq d(o_Y, \xi_t) - R - O(\delta).$$

Since $d(o_Y, \xi_t) = t$ and $d(o_Y, y) \leq \langle \xi, \eta \rangle_{o_Y} + O(\delta)$, this proves the desired estimate in this case.

If x is $R+O(\delta)$-close to $[o_Y, y]$, then choose $w \in \pi_{[o_Y,y]}(x)$. Then we have $d(x, w) < R+O(\delta)$. On the other hand, the concatenation $[w, y] \cup [y, x]$ is a $(1, O(\delta))$-quasi-geodesic. Hence y is also $R + O(\delta)$-close to x, and the same argument as above proves (1).

For (2), suppose that

$$\langle \xi, \eta \rangle_{o_Y} \geq t - R + O(\delta).$$

Then $\xi_{t-R+O(\delta)}$ is $O(\delta)$-close to $[o_Y, \eta]$. Hence ξ_t is R-close to $[o_Y, \eta]$, and therefore $\eta \in O_R(o_Y, \xi_t)$. □

Gromov models. For the rest of this section, let $(\Gamma, \mathcal{P})$ be a relatively hyperbolic group. Following [14], we recall the basic properties of Gromov models. Such a model is a proper geodesic Gromov hyperbolic space on which Γ acts in a way analogous to the action of a geometrically finite Kleinian group on the real hyperbolic space.

Let (Y, d) be a proper geodesic Gromov hyperbolic space. For any $\eta \in \partial Y$, a horofunction based at η is obtained as follows: if $y_n \in Y$ is a sequence converging to η, then, after passing to a subsequence, there exist $t_n \to \infty$ and a function $h : Y \to \mathbb{R}$ such that

$$h(x) = \lim_{n\to\infty} d(x, y_n) - t_n,$$

uniformly on compact subsets. There exists a constant $C > 0$, depending only on Y, such that every such function satisfies

$$\big|\big(h(\ell_\eta(t_2)) - h(\ell_\eta(t_1))\big) - (t_1 - t_2)\big| \le C \quad \text{for all } t_1, t_2 \ge 0, \tag{3.2}$$

for every unit-speed geodesic ray $\ell_\eta : [0, \infty) \to Y$ asymptotic to η. By a horofunction at η, we mean any function satisfying (3.2) for every such ray.

A subset $H \subset Y$ is called a horoball at η if there exists a horofunction h at η such that

$$\{h \le 0\} \subset H \subset \{h \le 10C\}.$$

Then $\overline{H} \cap \partial Y = \{\eta\}$, and horoballs are uniformly quasi-convex.

Definition 3.4. A proper geodesic Gromov hyperbolic space (Y, d) is called a *Gromov model* for $(\Gamma, \mathcal{P})$ if:

(1) Γ acts properly discontinuously on Y by isometries;
(2) Y is taut, meaning that there exists $R > 0$ such that every point of Y lies within distance R of a bi-infinite geodesic;
(3) there exists a Γ-invariant collection $\mathcal{B} = \{B_i\}$ of disjoint open horoballs such that the stabilizer of each B_i in Γ is of the form $\gamma \mathsf{P} \gamma^{-1}$ for some $\gamma \in \Gamma$ and $\mathsf{P} \in \mathcal{P}$;
(4) the action of Γ on $Y - \bigcup_i B_i$ is cocompact.

By the uniqueness of the Bowditch boundary, the Gromov boundary of a Gromov model Y for $(\Gamma, \mathcal{P})$ is Γ-equivariantly homeomorphic to $\partial(\Gamma, \mathcal{P})$. By [4, Proposition 2.12], the action of Γ on ∂Y is a convergence group action.

For each $B \in \mathcal{B}$, write

$$\{p\} = \overline{B} \cap \partial Y.$$

We call p the basepoint of B and also write $B = B_p$. Given $\mathsf{P} \in \mathcal{P}^\Gamma$, we denote its basepoint by $\xi_{\mathsf{P}} \in \partial Y$.

These basepoints are precisely the parabolic limit points of $(\Gamma, \mathcal{P})$ in ∂Y. We shall use the following standard properties of parabolic subgroups, whose proofs are included for completeness.

Lemma 3.5. *Let $o_Y \in Y$, let $\mathsf{P} \in \mathcal{P}^\Gamma$, and set $\xi = \xi_{\mathsf{P}}$. For every $R > 0$, there exists a compact set $Q \subset \partial Y - \{\xi\}$ such that*

$$O_R(o_Y, go_Y) \subset gQ \quad \textit{for all but finitely many } g \in \mathsf{P}.$$

Proof. Fix $R > 0$ and suppose the claim fails. Let $Q_n \subset \partial Y - \{\xi\}$ be an increasing sequence of compact sets with

$$\bigcup_n Q_n = \partial Y - \{\xi\}.$$

Then there exist sequences $g_n \in \mathsf{P}$ and $\eta_n \in O_R(o_Y, g_n o_Y)$ such that $\eta_n \notin g_n Q_n$. We may assume that the sequence g_n is infinite. Since $g_n^{-1}\eta_n \notin Q_n$ for every n, $g_n^{-1}\eta_n \to \xi$. On the other hand,

$$g_n^{-1}\eta_n \in O_R(g_n^{-1}o_Y, o_Y) \quad \text{and} \quad g_n^{-1}o_Y \to \xi,$$

which is impossible. This proves the lemma. □

Lemma 3.6. *Let $o_Y \in Y$, let $\mathsf{P} \in \mathcal{P}^\Gamma$, and set $\xi = \xi_{\mathsf{P}}$. For every compact set $Q \subset \partial Y - \{\xi\}$, there exists $R > 0$ such that*

$$gQ \subset O_R(o_Y, go_Y) \quad \textit{for all } g \in \mathsf{P}.$$

Proof. Let $Q \subset \partial Y - \{\xi\}$ be compact and suppose the conclusion fails. Then, for every $n \geq 1$, there exists $g_n \in \mathsf{P}$ such that

$$g_n Q \not\subset O_n(o_Y, g_n o_Y).$$

Thus g_n is an infinite sequence and

$$Q \not\subset O_n(g_n^{-1}o_Y, o_Y) \quad \text{for every } n \geq 1.$$

Since $g_n^{-1}o_Y \to \xi$, this forces $\xi \in Q$, a contradiction. □

Finally, a point $\xi \in \partial Y$ is conical if and only if, for any $o_Y \in Y$, there exist $R > 0$ and an infinite sequence $\gamma_n \in \Gamma$ such that

$$\xi \in O_R(o_Y, \gamma_n o_Y) \quad \text{for all } n.$$

In this case, we say that $\gamma_n o_Y$ converges conically to ξ. It is easy to see that this notion of conicality is equivalent to the conicality defined in terms of convergence action.

4. Relatively Morse groups

Let $\Gamma < G$ be a discrete subgroup which is hyperbolic relative to a finite collection $\mathcal{P}$ of finitely generated infinite subgroups of Γ. We fix a non-empty subset $\theta \subset \Pi$.

Definition 4.1. We say that Γ is *θ-Morse relative to $\mathcal{P}$* if there exist a Gromov model Y for $(\Gamma, \mathcal{P})$ and a Γ-equivariant quasi-isometric embedding

$$f : Y \to X$$

such that, if $\mathcal{L}_f \subset \mathfrak{a}^+$ denotes the asymptotic cone of

$$\{\mu(f(x)^{-1}f(y)) : x, y \in Y\},$$

then

$$\mathcal{L}_f \cap \ker \alpha = \{0\} \quad \text{for every } \alpha \in \theta.$$

Such a map f is called a Morse embedding of Y.

For a Morse embedding $f : Y \to X$, we call $\mathcal{L}_f$ the Morse limit cone of f. We also set

$$\mathcal{L}_{\theta,f} := p_\theta(\mathcal{L}_f)$$

and call it the Morse θ-limit cone. Since $\mathcal{L}_f = \mathrm{i}(\mathcal{L}_f)$, being θ-Morse relative to $\mathcal{P}$ is equivalent to being $\theta \cup \mathrm{i}(\theta)$-Morse relative to $\mathcal{P}$. Thus, without loss of generality, we assume

$$\theta = \mathrm{i}(\theta)$$

throughout the rest of this section.

A subgroup $\Gamma < G$ which is hyperbolic relative to $\mathcal{P}$ is called *θ-Anosov relative to $\mathcal{P}$* if:

- Γ is θ-regular, meaning that
$$\min_{\alpha\in\theta} \alpha(\mu(\gamma_n)) \to \infty$$
for every infinite sequence $\gamma_n \in \Gamma$;
- there exists a transverse Γ-equivariant embedding
$$\zeta : \partial Y \to \mathcal{F}_\theta,$$
i.e., it sends distinct points of ∂Y to points in general position.

If Γ is θ-Morse relative to $\mathcal{P}$, then Γ is θ-Anosov relative to $\mathcal{P}$. Indeed, θ-regularity follows from the fact that f is a quasi-isometric embedding and

$$\mathcal{L}_f \cap \bigcup_{\alpha\in\theta} \ker \alpha = \{0\}.$$

Moreover, by [15, Theorem 1.4], the Morse embedding $f : Y \to X$ extends continuously to a transverse Γ-equivariant embedding

$$f : \partial Y \to \mathcal{F}_\theta.$$

Its image is the θ-limit set Λ_θ.

Remark 4.2. A map $f : Y \to X$ satisfying

$$\mathcal{L}_f \cap \bigcup_{\alpha\in\theta} \ker \alpha = \{0\}$$

is called uniformly θ-regular in [14]. Although this condition is a priori different from the original definition of a Morse embedding in [14], the Morse lemma [15] implies that the two notions are equivalent for quasi-isometric embeddings.

Geometric properties of relatively Morse groups. For the rest of this section, let $\Gamma < G$ be θ-Morse relative to $\mathcal{P}$, with Morse embedding $f : Y \to X$ from a Gromov model (Y, d) and continuous extension $f : \partial Y \to \mathcal{F}_\theta$. We fix a basepoint $o_Y \in Y$ and we may assume that $f(o_Y) = o$.

The following proposition is a key ingredient in the proof of the global shadow lemma. Its second assertion follows from Lemma 4.4 below.

Proposition 4.3. *Let $\mathsf{P} \in \mathcal{P}$ and let $p = \xi_{\mathsf{P}}$. For every compact subset $Q \subset \partial Y - \{p\}$, there exists a constant $c = c(Q) > 0$ such that*

$$\sup_{\xi \in Q,\, \gamma \in \mathsf{P}} \left\| \mathcal{G}^\theta(f(p), \gamma f(\xi)) - \frac{1}{2}\mu_\theta(\gamma) \right\| \leq c.$$

In particular, $\mu_\theta(\gamma) - \mathrm{i}\,\mu_\theta(\gamma)$ is uniformly bounded for all $\gamma \in \mathsf{P}$.

We first collect several lemmas. The following was proved in [19, Lemma 6.6] when Γ is Borel Anosov, using the Morse property of the orbit map of an Anosov group into X [15]. The same argument applies in our setting, replacing the Cayley graph of a hyperbolic group by the Gromov hyperbolic space Y.

Lemma 4.4. *There exists $C > 0$ such that, for any distinct $\xi, \eta \in \partial Y$,*

$$\sup_{z \in \pi_{\xi,\eta}(o_Y)} \left\| \mathcal{G}^\theta(f(\xi), f(\eta)) - \frac{1}{2}\left(\mu_\theta(f(z)) + \mathrm{i}\,\mu_\theta(f(z))\right) \right\| \leq C.$$

The following was proved in [12, Corollary 4.12]. Although the original statement applies to the values of a linear form on the Cartan projections, the same proof gives the following vector-valued version.

Lemma 4.5. *There exists $D_0 > 0$ such that, for any $x, z \in Y$ and any $y \in [x, z]$,*

$$\left\| \mu_\theta(f(x)^{-1} f(z)) - \left(\mu_\theta(f(x)^{-1} f(y)) + \mu_\theta(f(y)^{-1} f(z))\right) \right\| < D_0.$$

We also need the following elementary observation.

Lemma 4.6. *Let $\mathsf{P} \in \mathcal{P}$ and let $p = \xi_{\mathsf{P}}$. For every $R > 0$, there exists a finite subset $\mathsf{P}(R) \subset \mathsf{P}$ such that, for every $\gamma \in \mathsf{P} - \mathsf{P}(R)$, every geodesic $[\xi, p]$ with $\xi \in \partial Y - \{p\}$, and every $u \in B(o_Y, R) \cap [\xi, p]$, we have*

$$\pi_{\gamma[\xi,p]}(u) \subset \gamma[u, p]$$

where we choose $[u, p] \subset [\xi, p]$.

Proof. Suppose not. Then there exist $R > 0$, an infinite sequence $\gamma_n \in \mathsf{P}$, and sequences

$$\xi_n \in \partial Y - \{p\}, \quad u_n \in B(o_Y, R) \cap [\xi_n, p], \quad \text{and} \quad y_n \in \pi_{\gamma_n[\xi_n,p]}(u_n),$$

such that

$$y_n \notin \gamma_n[u_n, p]$$

where we choose $[u_n, p] \subset [\xi_n, p]$, for all $n \geq 1$.

For each n, choose $[y_n, p] \subset \gamma_n[\xi_n, p]$. By the assumption, $\gamma_n u_n \in [y_n, p]$. By Gromov hyperbolicity, the concatenation

$$[u_n, y_n] \cup [y_n, p]$$

is a $(1, O(\delta))$-quasi-geodesic. Hence

$$p \in O_{O(\delta)}(u_n, \gamma_n u_n).$$

Since $d(o_Y, u_n) < R$, it follows that for all n,

$$p \in O_{R'}(o_Y, \gamma_n o_Y)$$

for some $R' > 0$ depending only on R and δ. Thus $\gamma_n o_Y$ converges conically to p, contradicting the fact that p is a parabolic limit point. □

We will use the following coarse midpoint estimate in the Gromov model:

Lemma 4.7. [6, Lemma 3.8] *Let $\mathsf{P} \in \mathcal{P}$ and let $p = \xi_{\mathsf{P}}$. Let $R > 0$, and let $Q \subset \partial Y - \{p\}$ be compact. Then there exists $D > 0$ such that, for every $x \in B(o_Y, R)$ and every $\gamma \in \mathsf{P}$,*

$$\sup_{\xi \in Q,\, y \in \pi_{\gamma\xi, p}(x)} \left| d(x, y) - \frac{1}{2} d(x, \gamma x) \right| \le D.$$

As an intermediate step in the proof of Proposition 4.3, we prove the following. We write $\approx$ for equality up to a uniform additive error.

Lemma 4.8. *Let $\mathsf{P} \in \mathcal{P}$ and let $p = \xi_{\mathsf{P}}$. Let $R > 0$, and let $Q \subset \partial Y - \{p\}$ be compact. Then there exists a constant $C = C(R, Q) > 0$ such that, for every $\gamma \in \mathsf{P}$, every geodesic $[\xi, p]$ with $\xi \in Q$, every $x \in [\xi, p] \cap B(o_Y, R)$, and every $w \in \gamma^{-1}\pi_{\gamma[\xi,p]}(x)$, there exists $y \in [x, \gamma x]$ such that*

$$d(y, w) < C, \quad d(y, \gamma w) < C, \quad \text{and} \quad d(w, \gamma w) < C.$$

Proof. Let $\mathsf{P}(R) \subset \mathsf{P}$ be the finite subset given by Lemma 4.6. It suffices to consider $\gamma \in \mathsf{P} - \mathsf{P}(R)$. Let $\xi \in Q$, and choose a geodesic $[\xi, p]$. Let $x \in [\xi, p] \cap B(o_Y, R)$, and let $w \in [\xi, p]$ be such that $\gamma w \in \pi_{\gamma[\xi,p]}(x)$. By Lemma 4.7,

$$d(x, \gamma w) \approx \frac{1}{2} d(x, \gamma x), \tag{4.1}$$

with additive error depending only on R and Q.

Consider the geodesic $[\gamma x, p] \subset \gamma[\xi, p]$ and the geodesic triangle

$$[x, p] \cup [x, \gamma x] \cup [\gamma x, p].$$

By Lemma 4.6, we have $\gamma w \in [\gamma x, p]$. Since $w \in [\xi, p]$, we also have $w \in [x, p]$. Since $\gamma w \in \pi_{\gamma[\xi,p]}(x)$, Gromov hyperbolicity implies that there exist $y \in [x, \gamma x]$ and $z \in [x, p]$ such that the three points γw, y, and z are uniformly close to one another. Together with (4.1), this gives

$$\frac{1}{2} d(x, \gamma x) \approx d(x, \gamma w) \approx d(x, y) \approx d(x, z).$$

Hence

$$d(x,\gamma x) \approx d(x,y) + d(x,z). \tag{4.2}$$

On the other hand,

$$\begin{aligned} d(x,\gamma x) &= d(x,y) + d(y,\gamma x) \\ &\approx d(x,y) + d(\gamma w, \gamma x) \\ &= d(x,y) + d(w,x). \end{aligned}$$

Comparing this with (4.2), we obtain

$$d(x,w) \approx d(x,z).$$

Since $z, w \in [x,p]$, this implies that the points z and w are uniformly close. Since γw, y, and z are uniformly close, it follows that γw, y, and w are uniformly close as well. This proves the lemma. □

Proof of Proposition 4.3. There exists $R = R(Q) > 0$ such that every geodesic $[\xi, p]$ with $\xi \in Q$ intersects $B(o_Y, R)$. Let $\gamma \in \mathsf{P}$.

Let $\xi \in Q$, and choose a geodesic $[\xi, p]$. Choose $x \in [\xi, p] \cap B(o_Y, R)$, and let $w \in [\xi, p]$ be such that

$$\gamma w \in \pi_{\gamma[\xi,p]}(x).$$

Since $x \in B(o_Y, R)$, the sets $\pi_{\gamma[\xi,p]}(x)$ and $\pi_{\gamma[\xi,p]}(o_Y)$ have uniformly bounded Hausdorff distance. Hence, by Lemmas 2.1 and 4.4,

$$\mathcal{G}^\theta(f(p), \gamma f(\xi)) \approx \frac{1}{2}\left(\mu_\theta(f(\gamma w)) + \mathrm{i}\,\mu_\theta(f(\gamma w))\right).$$

By Lemma 4.8, there exists $y \in [x, \gamma x]$ such that y, w, and γw are uniformly close. Since $y \in [x, \gamma x]$, Lemma 4.5 gives

$$\mu_\theta(f(x)^{-1} f(\gamma x)) \approx \mu_\theta(f(x)^{-1} f(y)) + \mu_\theta(f(y)^{-1} f(\gamma x)).$$

Since f is a quasi-isometric embedding, the points $f(y)$, $f(w)$, and $f(\gamma w)$ are uniformly close. Since $x \in B(o_Y, R)$, the point $f(x)$ remains in a uniformly bounded subset of X. Therefore, by Lemma 2.1,

$$\begin{aligned} \mu_\theta(\gamma) &\approx \mu_\theta(f(x)^{-1} f(\gamma x)) \\ &\approx \mu_\theta(f(x)^{-1} f(\gamma w)) + \mu_\theta(f(w)^{-1} f(x)) \\ &\approx \mu_\theta(f(x)^{-1} f(\gamma w)) + \mathrm{i}\,\mu_\theta(f(x)^{-1} f(w)) \\ &\approx \mu_\theta(f(\gamma w)) + \mathrm{i}\,\mu_\theta(f(\gamma w)) \\ &\approx 2\mathcal{G}^\theta(f(p), \gamma f(\xi)). \end{aligned}$$

This finishes the proof of the main claim.

For the second assertion, note that by Lemma 4.4,

$$\mathcal{G}^\theta(f(p), \gamma f(\xi)) \approx \frac{1}{2}\left(\mu_\theta(f(z_\gamma)) + \mathrm{i}\,\mu_\theta(f(z_\gamma))\right)$$

for some $z_\gamma \in \pi_{p,\gamma\xi}(o_Y)$. Hence

$$\mu_\theta(\gamma) \approx \mu_\theta(f(z_\gamma)) + \mathrm{i}\,\mu_\theta(f(z_\gamma)).$$

Since the right-hand side is i-invariant, this implies that $\mu_\theta(\gamma) - \mathrm{i}\,\mu_\theta(\gamma)$ is uniformly bounded. □

A metric-like function on Y. We identify $\mathfrak{a}_\theta^*$ with the subspace of $\mathfrak{a}^*$ obtained by precomposing with the projection $p_\theta : \mathfrak{a} \to \mathfrak{a}_\theta$. Let $\psi \in \mathfrak{a}_\theta^*$ be such that

$$\psi > 0 \quad \text{on } \mathcal{L}_f - \{0\}.$$

Define

$$\mathsf{d}_\psi(x, y) := \psi(\mu(f(x)^{-1} f(y))) \quad \text{for } x, y \in Y.$$

It was shown in [12] that d_ψ behaves like a metric on Y; for example, it satisfies a coarse triangle inequality. The following is a consequence of Lemma 4.5 and the Gromov hyperbolicity of Y. Since $\psi > 0$ on $\mathcal{L}_f - \{0\}$, there exists $c > 0$ such that

$$\mathsf{d}_\psi(x, y) > -c \quad \text{for all } x, y \in Y. \tag{4.3}$$

Proposition 4.9. *There exists $C > 0$ such that, for all $x, y, z \in Y$,*

$$\mathsf{d}_\psi(x, z) \le \mathsf{d}_\psi(x, y) + \mathsf{d}_\psi(y, z) + C.$$

Proof. Let $w \in \pi_{[x,z]}(y)$ be a nearest-point projection of y to a geodesic $[x, z]$. By Lemma 4.5,

$$\mathsf{d}_\psi(x, z) \le \mathsf{d}_\psi(x, w) + \mathsf{d}_\psi(w, z) + D_0 \|\psi\|,$$

where D_0 is the constant from Lemma 4.5. Since the concatenation $[x, w] \cup [w, y]$ is a uniform quasi-geodesic, Lemmas 3.1 and 4.5 imply that

$$\mathsf{d}_\psi(x, y) \approx \mathsf{d}_\psi(x, w) + \mathsf{d}_\psi(w, y) > \mathsf{d}_\psi(x, w) - c,$$

where c is as in (4.3). Similarly,

$$\mathsf{d}_\psi(y, z) \approx \mathsf{d}_\psi(y, w) + \mathsf{d}_\psi(w, z) > \mathsf{d}_\psi(w, z) - c.$$

Combining these three estimates gives the desired inequality. □

We also note that the metric-like function d_ψ behaves quasi-isometrically to the Gromov model (Y, d) as follows:

Lemma 4.10. (1) *For any $x, y, z \in Y$, if $y \in [x, z]$, then*

$$\mathsf{d}_\psi(x, z) \approx \mathsf{d}_\psi(x, y) + \mathsf{d}_\psi(y, z).$$

(2) *There exist $a, b > 0$ such that for all $x, y \in Y$,*

$$a\, d(x, y) - b \le \mathsf{d}_\psi(x, y) \le b\, d(x, y) + b.$$

(3) *There exists $A > 0$ such that, for every $\xi \in \partial Y$ and every $s \ge 0$, there exists $x \in [o_Y, \xi]$ such that*

$$|\mathsf{d}_\psi(o_Y, x) - s| \le A.$$

Proof. (1) is an immediate consequence of Lemma 4.5.

Since $\psi > 0$ on $\mathcal{L}_f - \{0\}$ and f is a quasi-isometric embedding, there exist $a, b > 0$ such that

$$a\, d(x,y) - b \le \mathsf{d}_\psi(x,y) \le b\, d(x,y) + b \tag{4.4}$$

for all $x, y \in Y$. This shows (2).

Finally, fix $\xi \in \partial Y$ and denote by $\xi_t \in [o_Y, \xi]$ the point such that $d(o_Y, \xi_t) = t$ for $t \ge 0$. We then set

$$F(t) := \mathsf{d}_\psi(o_Y, \xi_t).$$

Note that definition of $F(t)$ involves a choice of $[o_Y, \xi]$, but different choices only make uniformly bounded error which is allowed for our purpose. By Lemma 4.5, if $0 \le t \le u$, then

$$F(u) \approx F(t) + \mathsf{d}_\psi(\xi_t, \xi_u).$$

Together with (4.4), this gives constants $a_1, b_1 > 0$, independent of ξ, such that for all $0 \le t \le u$,

$$a_1(u-t) - b_1 \le F(u) - F(t) \le b_1(u-t) + b_1. \tag{4.5}$$

Since $F(0) = 0$ and $F(t) \to \infty$ as $t \to \infty$, for any $s \ge 0$ choose the smallest integer $n \ge 0$ with $F(n) \ge s$. If $n = 0$, then $s = 0$. If $n \ge 1$, then $F(n-1) < s$, and hence by (4.5),

$$0 \le F(n) - s \le F(n) - F(n-1) \le 2b_1.$$

Thus (3) holds with $A = 2b_1$. □

For subsets $E, F \subset Y$, set

$$\mathsf{d}_\psi(E,F) := \inf_{x \in E,\, y \in F} \mathsf{d}_\psi(x,y).$$

Lemma 4.11. *Let $\mathsf{P} \in \mathcal{P}$ and let $p = \xi_{\mathsf{P}}$. For every $x \in [o_Y, p]$, we have*

$$\mathsf{d}_\psi(x, \Gamma o_Y) \approx \mathsf{d}_\psi(x, o_Y),$$

where the implied constant depends only on the ambient constants and ψ.

Proof. Let $B_p \in \mathcal{B}$ be the horoball based at p. We first claim that

$$\mathsf{d}_\psi(x, \Gamma o_Y) \approx \mathsf{d}_\psi(x, \mathsf{P} o_Y).$$

It suffices to consider the case that x is sufficiently deep in B_p.

Since P acts cocompactly on ∂B_p, there exists $A > 0$ such that every point of ∂B_p is within distance A of some point of $\mathsf{P}o_Y$. Let $h \in \Gamma$. A geodesic from x to ho_Y leaves B_p through a point $u_h \in \partial B_p$. Choose $\gamma_h \in \mathsf{P}$ such that

$$d(u_h, \gamma_h o_Y) \le A.$$

By Lemma 2.1,

$$\mathsf{d}_\psi(x, \gamma_h o_Y) \approx \mathsf{d}_\psi(x, u_h).$$

On the other hand, since $u_h \in [x, ho_Y]$, Lemma 4.5 and (4.3) give

$$\mathsf{d}_\psi(x, ho_Y) \approx \mathsf{d}_\psi(x, u_h) + \mathsf{d}_\psi(u_h, ho_Y) \ge \mathsf{d}_\psi(x, u_h) - c$$

where $c > 0$ is given in (4.3). Hence

$$\mathsf{d}_\psi(x, \gamma_h o_Y) \le \mathsf{d}_\psi(x, h o_Y) + c'$$

for some uniform constant $c' > 0$ independent of h and x. Since this holds for all $h \in \Gamma$ and $\mathsf{P} < \Gamma$, the claim follows.

By the claim, we can choose $\gamma \in \mathsf{P}$ such that $\mathsf{d}_\psi(x, \Gamma o_Y) \approx \mathsf{d}_\psi(x, \gamma o_Y)$. Let $w \in \pi_{[o_Y,p]}(\gamma o_Y)$. By tautness of Y and Lemma 4.6, we may apply Lemma 4.8 to obtain a point $y \in [o_Y, \gamma^{-1} o_Y]$ such that the points w, $\gamma^{-1}w$, and y are uniformly close. Since f is a quasi-isometric embedding, the points $f(w)$, $f(\gamma^{-1}w)$, and $f(y)$ are also uniformly close. Hence

$$\mu_\theta(f(w)^{-1}) \approx \mu_\theta(f(\gamma^{-1}w)^{-1}) = \mu_\theta(f(w)^{-1} f(\gamma o_Y)).$$

We distinguish two cases according to the relative positions of w and x on $[o_Y, p]$. First suppose that $w \in [o_Y, x]$. By Lemma 4.5,

$$\begin{aligned}\mu_\theta(f(x)^{-1}) &\approx \mu_\theta(f(x)^{-1} f(w)) + \mu_\theta(f(w)^{-1}) \\ &\approx \mu_\theta(f(x)^{-1} f(w)) + \mu_\theta(f(w)^{-1} f(\gamma o_Y)) \\ &\approx \mu_\theta(f(x)^{-1} f(\gamma o_Y)),\end{aligned}$$

since the concatenation $[x, w] \cup [w, \gamma o_Y]$ is a $(1, O(\delta))$-quasi-geodesic. Applying ψ, we get

$$\mathsf{d}_\psi(x, o_Y) \approx \mathsf{d}_\psi(x, \gamma o_Y) \approx \mathsf{d}_\psi(x, \Gamma o_Y),$$

as desired.

Now suppose that $x \in [o_Y, w]$. Again by Lemma 4.5,

$$\begin{aligned}\mu_\theta(f(x)^{-1} f(\gamma o_Y)) &\approx \mu_\theta(f(x)^{-1} f(w)) + \mu_\theta(f(w)^{-1} f(\gamma o_Y)) \\ &\approx \mu_\theta(f(x)^{-1} f(w)) + \mu_\theta(f(w)^{-1}) \\ &\approx \mu_\theta(f(x)^{-1} f(w)) + \mu_\theta(f(w)^{-1} f(x)) + \mu_\theta(f(x)^{-1}).\end{aligned}$$

Applying ψ gives

$$\begin{aligned}\mathsf{d}_\psi(x, \gamma o_Y) &\approx \mathsf{d}_\psi(x, w) + \mathsf{d}_\psi(w, x) + \mathsf{d}_\psi(x, o_Y) \\ &> \mathsf{d}_\psi(x, o_Y) - 2c,\end{aligned}$$

where the last inequality follows from (4.3). Since $\mathsf{d}_\psi(x, \gamma o_Y) \approx \mathsf{d}_\psi(x, \Gamma o_Y)$ and $o_Y \in \Gamma o_Y$, the desired estimate follows in this case as well. □

$\mathfrak{a}_\theta$-valued Gromov products in shadows. We now translate the elementary comparison between shadows and Gromov products in the Gromov hyperbolic space Y into a comparison involving $\mathfrak{a}_\theta$-valued Gromov products on the flag variety. The main input from the relatively Morse property is that nearest-point projections in Y coarsely control the higher-rank Gromov product $\mathcal{G}^\theta(f(\xi), f(\eta))$ through Cartan projections; see Lemma 4.4. Together with the coarse additivity of Cartan projections along geodesics in Y, Lemma 4.5, this gives the following analogue of Lemma 3.3. For Anosov groups, this comparison was proved in [12]; the same argument applies in the relatively Morse setting.

For $\psi \in \mathfrak{a}_\theta^*$, write

$$\bar{\psi} := \frac{\psi + \psi \circ \mathrm{i}}{2}.$$

Proposition 4.12. *Let $\psi \in \mathfrak{a}_\theta^*$ be such that $\psi > 0$ on $\mathcal{L}_f - \{0\}$.*

(1) *For every $R > 0$, there exists $c_1 > 0$ such that, for every $\xi \in \partial Y$ and every $z \in [o_Y, \xi]$, if $\eta \in O_R(o_Y, z) - \{\xi\}$, then*

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar{\psi}}(o_Y, z) - c_1.$$

(2) *For every $c_2 > 0$, there exists $R > 0$ such that, for every $\xi \in \partial Y$ and every $z \in [o_Y, \xi]$, if $\eta \in \partial Y - \{\xi\}$ satisfies*

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar{\psi}}(o_Y, z) + c_2,$$

then

$$\eta \in O_R(o_Y, z).$$

Proof. We first prove (1). Fix $R > 0$, $\xi \in \partial Y$, and $z \in [o_Y, \xi]$. Let $\eta \in O_R(o_Y, z) - \{\xi\}$ and $y \in \pi_{[\xi,\eta]}(o_Y)$. Let $[z, \xi] \subset [o_Y, \xi]$.

We claim that there exists $r = r(R) > 0$ such that

$$[z, \xi] \cap B(y, r) \neq \emptyset. \tag{4.6}$$

Since $\eta \in O_R(o_Y, z)$, Lemma 3.1 implies that the uniform quasi-geodesic

$$[o_Y, y] \cup [y, \eta]$$

intersects $B(z, R + O(\delta))$. Choose $w \in [o_Y, y] \cup [y, \eta]$ such that

$$d(z, w) < R + O(\delta).$$

We consider two cases, according to the position of w relative to y.

If $w \in [o_Y, y]$, then both

$$[w, y] \cup [y, \xi] \quad \text{and} \quad [w, z] \cup [z, \xi]$$

are uniform quasi-geodesics with the same endpoints. By Lemma 3.1, the distance from y to $[w, z] \cup [z, \xi]$ is bounded above by a uniform constant depending on R. Since $[w, z]$ has length at most $R + O(\delta)$, the claim follows in this case.

If $w \in [y, \eta]$, choose a geodesic $[w, \xi] \subset [\xi, \eta]$. Then $y \in [w, \xi]$. As in the previous case,

$$[z, w] \cup [w, \xi]$$

is a uniform quasi-geodesic. Hence, by Lemma 3.1, the point $y \in [w, \xi]$ lies in a uniform neighborhood of $[z, \xi]$. Now the claim follows.

By (4.6), there exists $u \in [z, \xi] \subset [o_Y, \xi]$ such that $d(y, u) < r$. Since f is a quasi-isometric embedding, Lemma 2.1 implies that

$$\mu(f(y)) \approx \mu(f(u)),$$

where the implied constant depends on r. By Lemma 4.4,

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \approx \mathsf{d}_{\bar{\psi}}(o_Y, y) \approx \mathsf{d}_{\bar{\psi}}(o_Y, u).$$

Since $u \in [z, \xi] \subset [o_Y, \xi]$, Lemma 4.5 gives

$$\mathsf{d}_{\bar{\psi}}(o_Y, u) \approx \mathsf{d}_{\bar{\psi}}(o_Y, z) + \mathsf{d}_{\bar{\psi}}(z, u).$$

By (4.3), the term $\mathsf{d}_{\bar{\psi}}(z, u)$ is bounded from below by a uniform constant. Combining these estimates, we obtain a constant $c_1 > 0$ such that

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar{\psi}}(o_Y, z) - c_1.$$

This proves (1).

We now prove (2). Let $c_2 > 0$, let $\xi \in \partial Y$, and let $z \in [o_Y, \xi]$. Suppose that $\eta \in \partial Y - \{\xi\}$ satisfies

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar{\psi}}(o_Y, z) + c_2.$$

We claim that there exists $r_0 = r_0(c_2) > 0$ such that, for every $y \in \pi_{[\xi,\eta]}(o_Y)$, the union

$$[o_Y, y] \cup [y, \eta]$$

intersects $B(z, r_0)$. Since $[o_Y, y] \cup [y, \eta]$ and $[o_Y, \eta]$ have uniformly bounded Hausdorff distance by Lemma 3.1, the claim implies that

$$\eta \in O_R(o_Y, z) \quad \text{for } R = r_0 + O(\delta).$$

Let C be the constant from Lemma 4.4. Then

$$\mathsf{d}_{\bar{\psi}}(o_Y, y) \geq \mathsf{d}_{\bar{\psi}}(o_Y, z) + c_2 - \|\psi\| C. \tag{4.7}$$

By Lemma 3.1, there exists $w \in [o_Y, \xi]$ such that $d(y, w) < O(\delta)$. Since f is a quasi-isometric embedding,

$$\mathsf{d}_{\bar{\psi}}(o_Y, w) \approx \mathsf{d}_{\bar{\psi}}(o_Y, y). \tag{4.8}$$

We distinguish two cases according to the position of w relative to z. If $w \in [z, \xi] \subset [o_Y, \xi]$, then $z \in [o_Y, w]$. Since $d(y, w) < O(\delta)$, the concatenation

$$[o_Y, w] \cup [w, y]$$

is a $(1, O(\delta))$-quasi-geodesic. Hence, by Lemma 3.1, the geodesic $[o_Y, y]$ intersects a uniform neighborhood of z. Thus $[o_Y, y] \cup [y, \eta]$ intersects a uniform neighborhood of z.

If $w \in [o_Y, z] \subset [o_Y, \xi]$, then Lemma 4.5 gives

$$\mathsf{d}_{\bar{\psi}}(o_Y, z) \approx \mathsf{d}_{\bar{\psi}}(o_Y, w) + \mathsf{d}_{\bar{\psi}}(w, z).$$

Together with (4.7) and (4.8), this implies that $\mathsf{d}_{\bar{\psi}}(w, z)$ is uniformly bounded above. Since $\psi > 0$ on $\mathcal{L}_f - \{0\}$, the same is true for $\bar{\psi}$, and therefore $\|\mu(f(w)^{-1} f(z))\|$ is uniformly bounded. Since f is a quasi-isometric embedding, $d(w, z)$ is uniformly bounded. As $d(y, w) < O(\delta)$, it follows that $[o_Y, y] \cup [y, \eta]$ intersects a uniform neighborhood of z.

In both cases the claim follows, and this completes the proof. $\square$

5. Counting in parabolic subgroups

The counting estimates in this section require only the relatively Anosov hypothesis; no Morse embedding is assumed. The linear form ψ need not be invariant under opposition. They extend the mixed exponential growth estimates for parabolic subgroups in Hilbert geometry [6, Proposition 7.7] to Cartan lengths defined by arbitrary positive linear forms in semisimple Lie groups.

Throughout this section, let $\Gamma < G$ be θ-Anosov relative to a finite collection $\mathcal{P}$. Let $\mathcal{L}_\Gamma \subset \mathfrak{a}^+$ denote the limit cone of Γ, that is, the asymptotic cone of $\mu(\Gamma)$.

Fix $\psi \in \mathfrak{a}_\theta^*$ that is positive on $\mathcal{L}_\Gamma - \{0\}$. For a subgroup $H < \Gamma$, let $\delta_\psi(H)$ denote the abscissa of convergence of the Poincaré series

$$s \mapsto \sum_{\gamma \in H} e^{-s\psi(\mu(\gamma))}.$$

We prove the following counting estimate for parabolic subgroups.

Theorem 5.1. *Let $\mathsf{P} \in \mathcal{P}$. Then there exist $a_\psi(\mathsf{P}) \in \mathbb{Z}_{\geq 0}$, $C > 1$, $k \in \mathbb{N}$, and $T_0 > 0$ such that, for all $T > T_0$ and all $n \geq 0$,*

$$\begin{aligned} &C^{-1} e^{\delta_\psi(\mathsf{P})(T+kn)} (1+T+kn)^{a_\psi(\mathsf{P})} \\ &\leq \#\left\{\gamma \in \mathsf{P} : T + kn \leq \psi(\mu(\gamma)) < T + k(n+1)\right\} \\ &\leq C e^{\delta_\psi(\mathsf{P})(T+kn)} (1+T+kn)^{a_\psi(\mathsf{P})}. \end{aligned}$$

Moreover, if P is virtually cyclic or if G has rank one, then $a_\psi(\mathsf{P}) = 0$.

We deduce Theorem 5.1 from the following counting estimate. We write $\ll$ for inequality up to a uniform multiplicative constant, and similarly for $\gg$. We write $\asymp$ when we have both $\ll$ and $\gg$.

Proposition 5.2. *Let $\mathsf{P} \in \mathcal{P}$. Then there exist $a_\psi(\mathsf{P}) \in \mathbb{Z}_{\geq 0}$ and $c > 0$ such that, for all sufficiently large T,*

$$e^{\delta_\psi(\mathsf{P})T}(T-c)^{a_\psi(\mathsf{P})} \ll \#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) \leq T\} \ll e^{\delta_\psi(\mathsf{P})T}(T+c)^{a_\psi(\mathsf{P})}.$$

Moreover, if P is virtually cyclic or if G has rank one, then $a_\psi(\mathsf{P}) = 0$.

The key input is a volume estimate of Benoist-Oh [2]. We first recall the structural description of parabolic subgroups in the relatively Anosov setting. After the reduction in [9, Section 4.4], we may assume that P_θ contains no simple factor of G. Let $\mathsf{P} \in \mathcal{P}$. By [9, Theorem 4.4], there exists a closed subgroup $H < G$ with finitely many connected components such that P is a cocompact lattice in H. Moreover, if U denotes the unipotent radical of H, then

$$H = L \ltimes U \quad \text{and} \quad H^\circ = L^\circ \times U, \tag{5.1}$$

where $L < H$ is compact and its identity component L° is abelian.

Let $\mathfrak{u} = \mathrm{Lie}(U)$, and let ℓ denote Lebesgue measure on $\mathfrak{u}$. We shall use the following consequence of [2]:

Theorem 5.3. *Let $R_1, \dots, R_m$ be positive rational functions on $\mathfrak{u}$ which are defined everywhere. Let $c_1, \dots, c_m \in \mathbb{R}$, and suppose that $R := R_1^{c_1} \cdots R_m^{c_m}$ is a proper function. Then*

$$\ell\big(\{Y \in \mathfrak{u} : R(Y) \le T\}\big) \sim c_0 T^r (\log T)^q$$

for some $c_0 > 0$, $r \ge 0$, and $q \in \mathbb{Z}_{\ge 0}$.

Proof. This is immediate from [2, Proposition 7.2] when all exponents c_i are positive rational numbers. The same proof also gives the present form; we briefly indicate the additional points needed for real exponents. Following the notation in [2, Proposition 7.2], let Z be the affine space over $\mathbb{R}$ associated to the real vector space $\mathfrak{u}$, so that $Z(\mathbb{R}) = \mathfrak{u}$. Let ω be the standard algebraic volume form on Z inducing ℓ. Write $R_i = f_i/g_i$, with f_i and g_i polynomial. Choose a smooth projective real compactification V of Z. After replacing V by a resolution, and keeping the same notation, we may assume that $D := V - Z$, the divisors of the rational functions f_i, g_i, and the divisor of ω, have simple normal crossings. Put $F = R$ and $f = 1/F$.

Let $y_0 \in D(\mathbb{R}) \cap \overline{Z(\mathbb{R})}$. In a neighborhood of y_0, choose local real analytic coordinates $(x_1, \dots, x_N)$ such that

$$D = \{x_1 \cdots x_r = 0\}.$$

By the above resolution of singularities, and arguing as in the proof of [9, Lemma 6.7], we may write locally

$$f_i = x_1^{a_{i1}} \cdots x_r^{a_{ir}} \widehat{f}_i, \quad \text{and} \quad g_i = x_1^{b_{i1}} \cdots x_r^{b_{ir}} \widehat{g}_i,$$

where $a_{ij}, b_{ij} \in \mathbb{Z}$ and $\widehat{f}_i, \widehat{g}_i$ are nowhere-vanishing real analytic functions. After restricting to one orthant and replacing x_j by $|x_j|$, we may assume $x_j > 0$ for $j = 1, \dots, r$. Then $R_i = \widehat{R}_i \prod_{j=1}^r x_j^{a_{ij} - b_{ij}}$ where $\widehat{R}_i := \widehat{f}_i/\widehat{g}_i$, and hence

$$f = R^{-1} = \widehat{f} \prod_{j=1}^r x_j^{s_j}.$$

for some $s_j \in \mathbb{R}$ and some positive real analytic unit $\widehat{f}$. Since R is proper on $Z(\mathbb{R})$, $f = R^{-1}$ tends to 0 along every boundary component meeting the closure of $Z(\mathbb{R})$. Hence $s_j > 0$ for every relevant boundary component.

Moreover, locally

$$\omega = a_0(x) \prod_{j=1}^r x_j^{\beta_j} \, dx_1 \cdots dx_N$$

with $\beta_j \in \mathbb{Z}$ and a_0 a nowhere-vanishing real analytic function.

Thus the local integrals appearing in the proof of [2, Proposition 7.2] are of monomial type. The remainder of their argument applies without change. □

Proof of Proposition 5.2. By [9, Proposition 2.3], for each $\alpha \in \theta$ there exist $m_\alpha \in \mathbb{N}$, $C_\alpha > 1$, and an everywhere-defined positive rational function $R_\alpha : \mathfrak{u} \to \mathbb{R}$ such that, for all $Y \in \mathfrak{u}$,

$$C_\alpha^{-1} R_\alpha(Y)^{1/m_\alpha} \le e^{\omega_\alpha(\mu(\exp Y))} \le C_\alpha R_\alpha(Y)^{1/m_\alpha},$$

where $\omega_\alpha \in \mathfrak{a}_\theta^*$ is the fundamental weight associated to α.

Write

$$\psi = \sum_{\alpha\in\theta} c_\alpha \omega_\alpha$$

for coefficients $c_\alpha \in \mathbb{R}$, $\alpha \in \theta$. Set

$$R_\psi := \prod_{\alpha\in\theta} R_\alpha^{c_\alpha/m_\alpha} \quad \text{and} \quad C_\psi := \prod_{\alpha\in\theta} C_\alpha^{|c_\alpha|}.$$

Then, for all $Y \in \mathfrak{u}$,

$$C_\psi^{-1} R_\psi(Y)^{-1} \le e^{-\psi(\mu(\exp Y))} \le C_\psi R_\psi(Y)^{-1}.$$

By [9, Lemma 7.3], the function R_ψ is proper.

Let

$$\mathsf{P}_1 := \pi(\mathsf{P} \cap H^\circ),$$

where $\pi : H^\circ \to U$ is the projection. Then P_1 is a cocompact lattice in U. Since $\mathsf{P} \cap H^\circ$ has finite index in P and $\ker \pi$ is compact, Lemma 2.1 gives a constant $c > 0$ such that

$$\begin{aligned}\#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T - c\} &\ll \#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) \le T\} \\ &\ll \#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T + c\}.\end{aligned}$$

It therefore suffices to prove the stated growth estimate for P_1, since bounded shifts of T do not change that estimate.

Let λ_U denote a Haar measure on U. Let $Q_1 \subset U$ be a bounded open set such that the translates gQ_1, $g \in \mathsf{P}_1$, are pairwise disjoint. Then

$$\begin{aligned}\#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T\} &\ll \lambda_U \left(\bigcup_{\substack{g\in\mathsf{P}_1 \\ \psi(\mu(g))\le T}} gQ_1 \right) \\ &\ll \lambda_U \left(\{u \in U : \psi(\mu(u)) \le T + c'\} \right) \\ &\ll \ell \left(\{Y \in \mathfrak{u} : R_\psi(Y) \le c e^T\} \right)\end{aligned}$$

for some constants $c', c > 1$.

Similarly, choose a compact set $Q_2 \subset U$ such that $\mathsf{P}_1 Q_2 = U$. Then, after increasing c', c with $c > 1$, if necessary,

$$\begin{aligned}\#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T\} &\gg \lambda_U \left(\bigcup_{\substack{g\in \mathsf{P}_1 \\ \psi(\mu(g))\le T}} gQ_2 \right) \\ &\gg \lambda_U \left(\{u \in U : \psi(\mu(u)) \le T - c'\} \right) \\ &\gg \ell \left(\{Y \in \mathfrak{u} : R_\psi(Y) \le c^{-1} e^T\} \right).\end{aligned}$$

The proposition now follows from Theorem 5.3, together with the fact that the exponential growth rate of

$$\#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T\}$$

is $\delta_\psi(\mathsf{P})$. If P is virtually cyclic, then the corresponding unipotent group is one-dimensional, and the volume asymptotic has no logarithmic factor. Hence $a_\psi(\mathsf{P}) = 0$. See Theorem 5.5 for the rank one case. □

Before proving Theorem 5.1, we record the following entropy gap for parabolic subgroups, due to Canary–Zhang–Zimmer.

Theorem 5.4. [9, Lemma 7.4, Theorem 6.1, Theorem 7.1] *For each $\mathsf{P} \in \mathcal{P}$, we have*

$$0 < \delta_\psi(\mathsf{P}) < \delta_\psi(\Gamma).$$

Proof of Theorem 5.1. By Proposition 5.2, there exist $c > 0$ and $T_0 > 0$ such that, for all $T > T_0$,

$$e^{\delta_\psi(\mathsf{P})T - c}(1+T)^{a_\psi(\mathsf{P})} \le \#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) < T\} \le e^{\delta_\psi(\mathsf{P})T + c}(1+T)^{a_\psi(\mathsf{P})}.$$

For simplicity, write

$$\delta = \delta_\psi(\mathsf{P}) \quad \text{and} \quad a = a_\psi(\mathsf{P}),$$

and define

$$N(T) := \#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) < T\}.$$

Choose $k \in \mathbb{N}$ so large that

$$e^{\delta k - c} - e^c > 0,$$

which is possible since $\delta > 0$ by Theorem 5.4. Set $S := T + kn$. Then

$$\begin{aligned}N(S+k) - N(S) &\le e^{\delta(S+k)+c}(1+S+k)^a - e^{\delta S - c}(1+S)^a \\ &= e^{\delta S}(1+S)^a \left[e^{\delta k + c} \left(1 + \frac{k}{1+S}\right)^a - e^{-c} \right] \\ &\le C e^{\delta S}(1+S)^a\end{aligned}$$

for a constant $C > 1$ independent of T and n. Similarly,

$$\begin{aligned} N(S+k) - N(S) &\geq e^{\delta(S+k)-c}(1+S+k)^a - e^{\delta S+c}(1+S)^a \\ &= e^{\delta S}(1+S)^a \left[e^{\delta k - c} \left(1 + \frac{k}{1+S}\right)^a - e^c \right] \\ &\geq C^{-1} e^{\delta S}(1+S)^a, \end{aligned}$$

after increasing C, if necessary. Since

$$N(S+k) - N(S) = \#\{\gamma \in \mathsf{P} : S \leq \psi(\mu(\gamma)) < S+k\},$$

and $S = T + kn$, this proves the theorem. □

Rank-one parabolic subgroups. We record a rank-one refinement of Proposition 5.2, which shows the absence of the polynomial term in Theorem 5.1 for rank-one G. In this subsection, assume that G has real rank one. Let $\Pi = \{\alpha\}$, so that $\theta = \{\alpha\}$ and $\mathfrak{a}_\theta = \mathfrak{a} \simeq \mathbb{R}$. Let $\mathfrak{n} := \mathrm{Lie}(N)$ and write

$$\mathfrak{n} = \mathfrak{g}_\alpha \oplus \mathfrak{g}_{2\alpha},$$

where $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{2\alpha}$ are the corresponding root spaces, with the convention that $\mathfrak{g}_{2\alpha} = 0$ if 2α is not a root.

Let $\mathsf{P} \in \mathcal{P}$. Let $H = L \ltimes U$ be the subgroup associated to P as in (5.1). Up to conjugation, we may assume that $U < N$, and use the same notation $\mathfrak{u} := \mathrm{Lie}(U)$ as before. Define

$$V_\mathsf{P} := \mathrm{pr}_{\mathfrak{g}_\alpha}(\mathfrak{u}), \quad Z_\mathsf{P} := \mathfrak{u} \cap \mathfrak{g}_{2\alpha}, \quad \text{and} \quad Q(\mathsf{P}) := \dim V_\mathsf{P} + 2 \dim Z_\mathsf{P}.$$

Since $\mathfrak{a}$ is one-dimensional, $\mathcal{L}_\Gamma = \mathfrak{a}^+$ and hence any $\psi \in \mathfrak{a}^*$ positive on $\mathcal{L}_\Gamma - \{0\}$ is multiplication by a positive real number. The norm $\|\psi\|$ is given by $\psi(H_0)$ where $H_0 \in \mathfrak{a}^+$ is the unique unit vector.

Theorem 5.5. *Suppose that rank $G = 1$. For any positive $\psi \in \mathfrak{a}^*$ on $\mathfrak{a}^+ - \{0\}$, we have*

$$\#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) \leq T\} \asymp e^{\delta_\psi(\mathsf{P})T},$$

and $\delta_\psi(\mathsf{P}) = \frac{\|\alpha\| Q(\mathsf{P})}{2\|\psi\|}$. In particular, in Theorem 5.1, we have

$$a_\psi(\mathsf{P}) = 0 = a_{\bar{\psi}}(\mathsf{P}).$$

Proof. Let $H_0 \in \mathfrak{a}^+$ be the unit vector. Let $b := \alpha(H_0)$ and $\lambda_\psi := \psi(H_0)$. We use the same reduction as in Proposition 5.2. Recall that $\pi : H^\circ \to U$ is the projection and $\mathsf{P}_1 = \pi(\mathsf{P} \cap H^\circ)$. Then P_1 is a cocompact lattice in U. Since $\mathsf{P} \cap H^\circ$ has finite index in P and $\ker \pi$ is compact, Lemma 2.1 implies that counting P and counting P_1 give the same estimates, up to multiplicative constants, as before.

Write $u = \exp(X + Z) \in U$ with $X \in \mathfrak{g}_\alpha$ and $Z \in \mathfrak{g}_{2\alpha}$. By [20, (2.5)], following [23, p. 72],

$$\cosh^2\left(\frac{d_{NA}(o, uo)}{2}\right) = \left(1 + \frac{\|X\|^2}{8}\right)^2 + \frac{1}{4}\|Z\|^2$$

where d_{NA} is the distance induced from the norm whose unit vector H_1 satisfies $\alpha(H_1) = 1/2$. It follows that d_{NA} is the $2b$ multiple of the Riemannian distance d induced from the norm for which H_0 is a unit vector.

Therefore

$$d(o, uo) \approx \frac{2}{b} \log \max\{1, \|X\|, \|Z\|^{1/2}\}.$$

Define

$$\|u\|_{\mathrm{cusp}} := \max\{1, \|X\|, \|Z\|^{1/2}\},$$

where the term involving Z is omitted if $\mathfrak{g}_{2\alpha} = 0$. Since $\mu(u) = d(o, uo)H_0$, we get

$$\psi(\mu(u)) = \lambda_\psi d(o, uo) \approx \frac{2\lambda_\psi}{b} \log\|u\|_{\mathrm{cusp}}.$$

Hence

$$\psi(\mu(u)) \le T \quad \Longleftrightarrow \quad \|u\|_{\mathrm{cusp}} \ll e^{bT/(2\lambda_\psi)},$$

up to changing the implicit constants.

It remains to compute the volume growth of these $\|\cdot\|_{\mathrm{cusp}}$-balls inside U. Choose a linear complement W to Z_{P} in $\mathfrak{u}$. The projection $W \to V_{\mathsf{P}}$ is an isomorphism, so every $Y \in \mathfrak{u}$ can be written uniquely as

$$Y = v + \phi(v) + z \quad \text{with } v \in V_{\mathsf{P}} \text{ and } z \in Z_{\mathsf{P}},$$

for some linear map $\phi : V_{\mathsf{P}} \to \mathfrak{g}_{2\alpha}$. The condition

$$\|\exp Y\|_{\mathrm{cusp}} \le R$$

is equivalent, up to uniform constants, to

$$\|v\| \ll R \quad \text{and} \quad \|\phi(v) + z\| \ll R^2.$$

Thus the v-variables contribute $R^{\dim V_{\mathsf{P}}}$ and the z-variables contribute $R^{2\dim Z_{\mathsf{P}}}$. Therefore

$$\lambda_U(\{u \in U : \|u\|_{\mathrm{cusp}} \le R\}) \asymp R^{Q(\mathsf{P})}.$$

Since P_1 is a cocompact lattice in U, a compact fundamental domain comparison gives

$$\#\{g \in \mathsf{P}_1 : \|g\|_{\mathrm{cusp}} \le R\} \asymp R^{Q(\mathsf{P})}.$$

Substituting $R = e^{bT/(2\lambda_\psi)}$ yields

$$\#\{g \in \mathsf{P}_1 : \psi(\mu(g)) \le T\} \asymp e^{\frac{bQ(\mathsf{P})}{2\lambda_\psi} T}.$$

This finishes the proof. Note that the last claim follows since i is trivial in rank one. □

6. Shadow estimates at parabolic limit points

Let $\Gamma < G$ be θ-Morse relative to $\mathcal{P}$, with Morse embedding $f : Y \to X$ of a Gromov model (Y, d) for $(\Gamma, \mathcal{P})$. Let $\psi \in \mathfrak{a}_\theta^*$ be positive on $\mathcal{L}_f - \{0\}$. We normalize ψ so that $\delta_\psi(\Gamma) = 1$. Then the existence and uniqueness of the Patterson–Sullivan measure is a consequence of the work of Canary–Zhang–Zimmer.

Theorem 6.1. [9] *There exists a unique (Γ, ψ)-Patterson–Sullivan measure ν on Λ_θ. Moreover, ν is atomless.*

Let ν be a (Γ, ψ)-Patterson–Sullivan measure on Λ_θ given in Theorem 6.1. In this section we prove the global shadow lemma for shadows along geodesic rays towards parabolic limit points and provide estimates on their complements. Since all shadows we consider are taken in $Y \cup \partial Y$, we identify ν with its pullback to ∂Y under the Γ-equivariant homeomorphism $f : \partial Y \to \Lambda_\theta$.

Suppose $\theta = \mathrm{i}(\theta)$ and set

$$\bar{\psi} := \frac{\psi + \psi \circ \mathrm{i}}{2}.$$

Since $\psi > 0$ on $\mathcal{L}_f - \{0\}$, so is $\bar{\psi}$, and hence there exists $C_{\bar{\psi}} > 0$ such that

$$\bar{\psi}(v) > 1 - C_{\bar{\psi}} \quad \text{for all } v \in \mu(f(Y)^{-1} f(Y)). \tag{6.1}$$

Shadows at parabolic limit points. We first estimate shadows along geodesic rays ending at parabolic limit points. Recall that $o_Y \in Y$ is chosen so that $f(o_Y) = o$. For $\xi \in \partial Y$ and $t \geq 0$, let $\xi_t \in [o_Y, \xi]$ denote the point with $d(o_Y, \xi_t) = t$, after choosing a geodesic ray $[o_Y, \xi]$.

Theorem 6.2. *Let $\mathsf{P} \in \mathcal{P}$ and let $\xi = \xi_{\mathsf{P}}$. For all sufficiently large $R > 0$,*

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{2(\delta_{\bar{\psi}}(\mathsf{P}) - 1)\mathsf{d}_{\bar{\psi}}(o_Y, \xi_t)} \big(C_{\bar{\psi}} + \mathsf{d}_{\bar{\psi}}(o_Y, \xi_t)\big)^{a_{\bar{\psi}}(\mathsf{P})}$$

for all $t \geq 0$, with implied constants independent of t.

For an arbitrary parabolic limit point, we obtain the following translated form.

Theorem 6.3. *There exist constants $c, R_0 > 0$ with the following property. Let $\mathsf{P} \in \mathcal{P}$ and let $\xi = \gamma \xi_{\mathsf{P}}$ for some $\gamma \in \Gamma$. Suppose that*

$$d(\xi_{t_0}, \gamma o_Y) < c \quad \text{for some } t_0 \geq 0.$$

Then, for every $C \geq C_{\bar{\psi}}$, $R > R_0$, and $t \geq t_0$,

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{-\psi(\mu(\gamma))} e^{2(\delta_{\bar{\psi}}(\mathsf{P}) - 1)\mathsf{d}_{\bar{\psi}}(\gamma o_Y, \xi_t)} \cdot \big(C + \mathsf{d}_{\bar{\psi}}(\gamma o_Y, \xi_t)\big)^{a_{\bar{\psi}}(\mathsf{P})}.$$

For $g \in G$, define the translated measure ν_g by

$$d\nu_g(\eta) = e^{\psi(\beta_\eta^\theta(e, g))}\, d\nu(\eta).$$

Then, for $g, h \in G$,

$$d\nu_g(\eta) = e^{\psi(\beta_\eta^\theta(h, g))}\, d\nu_h(\eta).$$

For $x \in Y$, we write

$$\nu_x := \nu_g$$

where $g \in G$ satisfies $f(x) = go$. This is independent of the choice of g.

The next lemma relates ν to its translate at the center of a shadow.

Lemma 6.4. *For each fixed $R > 0$, uniformly for $\xi \in \partial Y$ and $t \geq 0$,*

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{-\mathsf{d}_\psi(o_Y, \xi_t)} \nu_{\xi_t}(O_R(o_Y, \xi_t)).$$

Lemma 6.4 is an immediate consequence of the following comparison between Busemann maps and Cartan projections inside shadows. The corresponding statement for shadows in the symmetric space $X = G/K$ was proved in [19, Lemma 5.7]; the present version follows from comparing shadows in Y with shadows in X under the Morse embedding f.

Lemma 6.5. *For every $R > 0$, there exists $C > 0$ such that, for all $x, y \in Y$ and $\xi \in O_R(x, y)$,*

$$\left\| \beta^\theta_{f(\xi)}(g, h) - \mu_\theta(g^{-1}h) \right\| < C$$

whenever $g, h \in G$ satisfy $go = f(x)$ and $ho = f(y)$.

Since Γ acts cocompactly on $Y - \bigcup \mathcal{B}$, Lemma 6.4 and the ordinary shadow lemma imply the following thick-part estimate.

Lemma 6.6. *There exists $R_0 > 0$ such that, for every $R > R_0$ and every $x \in Y - \bigcup \mathcal{B}$,*

$$\nu_x(O_R(o_Y, x)) \asymp 1.$$

The main estimate needed for Theorems 6.2 and 6.3 is the following.

Proposition 6.7. *Let $\mathsf{P} \in \mathcal{P}$ and let $\xi = \xi_{\mathsf{P}}$. For all sufficiently large $R > 0$, there exists $C > 0$ such that, for all $t \geq 0$,*

$$\nu_{\xi_t}(O_R(o_Y, \xi_t)) \ll e^{\mathsf{d}_\psi(o_Y, \xi_t)} \sum_{\substack{g \in \mathsf{P} \\ \bar\psi(\mu(g)) \geq 2\mathsf{d}_{\bar\psi}(o_Y, \xi_t) - C}} e^{-\psi(\mu(g))},$$

$$\nu_{\xi_t}(O_R(o_Y, \xi_t)) \gg e^{\mathsf{d}_\psi(o_Y, \xi_t)} \sum_{\substack{g \in \mathsf{P} \\ \bar\psi(\mu(g)) \geq 2\mathsf{d}_{\bar\psi}(o_Y, \xi_t) + C}} e^{-\psi(\mu(g))}.$$

Proof. Choose $R > 0$ large enough so that the ordinary shadow lemma holds. Let $Q \subset \partial Y - \{\xi\}$ be compact such that $\mathsf{P}Q = \partial Y - \{\xi\}$ and such that the conclusion of Lemma 3.5 holds for the chosen R. Increasing R if necessary, Proposition 4.12 gives constants $c_1, c_2 > 0$ such that

$$\eta \in O_R(o_Y, \xi_t) - \{\xi\} \Longrightarrow \psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar\psi}(o_Y, \xi_t) - c_1,$$

$$\psi(\mathcal{G}^\theta(f(\xi), f(\eta))) \geq \mathsf{d}_{\bar\psi}(o_Y, \xi_t) + c_2 \Longrightarrow \eta \in O_R(o_Y, \xi_t).$$

By Proposition 4.3, and using that $\mu_\theta(g) - \mathrm{i}\,\mu_\theta(g)$ is uniformly bounded for $g \in \mathsf{P}$, there exists $c > 0$ such that

(6.2)
$$\bigcup_{\substack{g\in\mathsf{P}\\ \frac{1}{2}\bar\psi(\mu(g))\geq \mathsf{d}_{\bar\psi}(o_Y,\xi_t)+c+c_2}} gQ \subset O_R(o_Y,\xi_t) - \{\xi\} \subset \bigcup_{\substack{g\in\mathsf{P}\\ \frac{1}{2}\bar\psi(\mu(g))\geq \mathsf{d}_{\bar\psi}(o_Y,\xi_t)-c-c_1}} gQ.$$

If $gQ \subset O_R(o_Y,\xi_t)$, then Lemma 6.5 gives

$$\nu_{\xi_t}(gQ) \asymp e^{\mathsf{d}_\psi(o_Y,\xi_t)}\nu(gQ).$$

Similarly, if $gQ \cap O_R(o_Y,\xi_t) \neq \emptyset$, then

$$\nu_{\xi_t}(gQ \cap O_R(o_Y,\xi_t)) \ll e^{\mathsf{d}_\psi(o_Y,\xi_t)}\nu(gQ).$$

Since ν is atomless (Theorem 6.1), using (6.2) and the bounded multiplicity of the P-translates of Q, we obtain

$$\nu_{\xi_t}(O_R(o_Y,\xi_t)) \ll e^{\mathsf{d}_\psi(o_Y,\xi_t)} \sum_{\substack{g\in\mathsf{P}\\ \frac{1}{2}\bar\psi(\mu(g))\geq \mathsf{d}_{\bar\psi}(o_Y,\xi_t)-c-c_1}} \nu(gQ),$$

$$\nu_{\xi_t}(O_R(o_Y,\xi_t)) \gg e^{\mathsf{d}_\psi(o_Y,\xi_t)} \sum_{\substack{g\in\mathsf{P}\\ \frac{1}{2}\bar\psi(\mu(g))\geq \mathsf{d}_{\bar\psi}(o_Y,\xi_t)+c+c_2}} \nu(gQ).$$

By Lemmas 3.5 and 3.6, and the ordinary shadow lemma,

$$\nu(gQ) \asymp e^{-\psi(\mu(g))}.$$

The proposition follows. □

Proof of Theorem 6.2. Let $C > 0$ be the constant from Proposition 6.7. Let k and T_0 be as in Theorem 5.1, applied to $\bar\psi$.

First suppose that

$$2\mathsf{d}_{\bar\psi}(o_Y,\xi_t) - C > T_0.$$

Set

$$T := 2\mathsf{d}_{\bar\psi}(o_Y,\xi_t) - C.$$

By Proposition 4.3, $\psi(\mu(g)) \approx \bar\psi(\mu(g))$ for $g \in \mathsf{P}$. Hence $\delta_{\bar\psi}(\mathsf{P}) = \delta_\psi(\mathsf{P}) < 1$ by Theorem 5.4. Together with the entropy gap (Theorem 5.4), this implies

$\delta_{\bar\psi}(\mathsf{P}) < 1$. Hence Theorem 5.1 gives

$$\begin{aligned}
&\sum_{\substack{g\in\mathsf{P}\\ \bar\psi(\mu(g))\ge T}} e^{-\psi(\mu(g))}\\
&\quad\ll \sum_{n=0}^{\infty} e^{-(T+kn)} \#\left\{g\in\mathsf{P} : T+kn\le \bar\psi(\mu(g)) < T+k(n+1)\right\}\\
&\quad\ll \sum_{n=0}^{\infty} e^{(\delta_{\bar\psi}(\mathsf{P})-1)(T+kn)}(1+T+kn)^{a_{\bar\psi}(\mathsf{P})}\\
&\quad\asymp e^{(\delta_{\bar\psi}(\mathsf{P})-1)T}(1+T)^{a_{\bar\psi}(\mathsf{P})}\\
&\quad\asymp e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(o_Y,\xi_t)}\big(C_{\bar\psi}+\mathsf{d}_{\bar\psi}(o_Y,\xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.
\end{aligned}$$

The lower bound is proved in the same way. Thus by Proposition 6.7,

$$\nu_{\xi_t}(O_R(o_Y,\xi_t)) \asymp e^{\mathsf{d}_\psi(o_Y,\xi_t)} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(o_Y,\xi_t)} \cdot \big(C_{\bar\psi}+\mathsf{d}_{\bar\psi}(o_Y,\xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.$$

Applying Lemma 6.4 proves the theorem in this case.

It remains to consider the case when $2\mathsf{d}_{\bar\psi}(o_Y,\xi_t) - C \le T_0$. Since $\bar\psi$ is positive on $\mathcal{L}_f - \{0\}$, this implies that $d(o_Y,\xi_t)$ is uniformly bounded. Hence, after increasing R by a uniform amount, $O_R(o_Y,\xi_t) = \partial Y$, and the desired estimate is trivial. □

Proof of Theorem 6.3. Let $\xi = \gamma_0\xi_{\mathsf{P}}$ for some $\gamma_0\in\Gamma$, and let $B_\xi\in\mathcal{B}$ be the horoball based at ξ. Then $\mathrm{Stab}_\Gamma(B_\xi) = \gamma_0\mathsf{P}\gamma_0^{-1}$. Since $\gamma_0\mathsf{P}\gamma_0^{-1}$ acts cocompactly on ∂B_ξ, there exist a uniform constant $c>0$, an element $\gamma_1\in\mathsf{P}$, and $t_0\ge 0$ such that

$$d(\xi_{t_0}, \gamma_0\gamma_1 o_Y) < c.$$

Set $\gamma := \gamma_0\gamma_1$.

For $s\ge 0$, let $\eta_s\in[\gamma o_Y,\xi]$ be the point satisfying $d(\gamma o_Y,\eta_s) = s$. Increasing c by a uniform amount, if necessary, we have

$$d(\xi_t,\eta_{t-t_0}) < c \quad \text{for all } t\ge t_0.$$

Thus, there exists $c'>0$ so that for all sufficiently large R,

$$O_{R-c'}(\gamma o_Y,\eta_{t-t_0}) \subset O_R(o_Y,\xi_t) \subset O_{R+c'}(\gamma o_Y,\eta_{t-t_0}) \tag{6.3}$$

for all large $t\ge t_0$.

Moreover,

$$O_R(o_Y,\xi_t) \subset O_{R+c}(o_Y,\gamma o_Y).$$

Using Lemma 6.5, we obtain

$$\begin{aligned}\nu(O_R(o_Y,\xi_t)) &= \nu\big(\gamma\,\gamma^{-1}O_R(o_Y,\xi_t)\big)\\ &= \int_{\gamma^{-1}O_R(o_Y,\xi_t)} e^{\psi(\beta_x^\theta(e,\gamma^{-1}))}\,d\nu(x)\\ &= \int_{\gamma^{-1}O_R(o_Y,\xi_t)} e^{-\psi(\beta_{\gamma x}^\theta(e,\gamma))}\,d\nu(x)\\ &\asymp e^{-\psi(\mu(\gamma))}\nu(\gamma^{-1}O_R(o_Y,\xi_t)).\end{aligned}$$

By (6.3) and Theorem 6.2,

$$\nu(O_R(o_Y,\xi_t)) \asymp e^{-\psi(\mu(\gamma))}e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma o_Y,\eta_{t-t_0})}\cdot\big(C_{\bar\psi}+\mathsf{d}_{\bar\psi}(\gamma o_Y,\eta_{t-t_0})\big)^{a_{\bar\psi}(\mathsf{P})}.$$

Since (6.1) holds, the factor $C_{\bar\psi}+\mathsf{d}_{\bar\psi}$ may be replaced, up to multiplicative constants, by $C+\mathsf{d}_{\bar\psi}$ for any $C\geq C_{\bar\psi}$. Finally, ξ_t and η_{t-t_0} are uniformly close, so Lemma 2.1 gives

$$\mathsf{d}_{\bar\psi}(\gamma o_Y,\eta_{t-t_0})\approx\mathsf{d}_{\bar\psi}(\gamma o_Y,\xi_t).$$

This proves the theorem. □

Complements of parabolic shadows. We also need an estimate for the complement of a shadow based at a parabolic limit point.

Proposition 6.8. *Let $\xi=\xi_{\mathsf{P}}$ for some $\mathsf{P}\in\mathcal{P}$. For all sufficiently large $R>0$ and all sufficiently large $t\geq 0$,*

$$\nu_{\xi_t}(\partial Y-O_R(o_Y,\xi_t)) \asymp e^{-\mathsf{d}_\psi(\xi_t,o_Y)}e^{2\delta_{\bar\psi}(\mathsf{P})\mathsf{d}_{\bar\psi}(o_Y,\xi_t)}\cdot\big(C_{\bar\psi}+\mathsf{d}_{\bar\psi}(o_Y,\xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.$$

Proof. Let $R>0$ and $Q\subset\partial Y-\{\xi\}$ be as in the proof of Proposition 6.7. For every $g\in\mathsf{P}$ and $\eta\in gQ$, Proposition 4.3 gives

$$\Big|\psi(\mathcal{G}^\theta(f(\xi),f(\eta)))-\tfrac12\bar\psi(\mu(g))\Big|\leq c.$$

Together with Proposition 4.12 and $\mathsf{P}Q=\partial Y-\{\xi\}$, this implies

$$\bigcup_{\substack{g\in\mathsf{P}\\ \frac12\bar\psi(\mu(g))<\mathsf{d}_{\bar\psi}(o_Y,\xi_t)-c-c_1}} gQ\subset\partial Y-O_R(o_Y,\xi_t)\subset\bigcup_{\substack{g\in\mathsf{P}\\ \frac12\bar\psi(\mu(g))<\mathsf{d}_{\bar\psi}(o_Y,\xi_t)+c+c_2}} gQ.$$

The thin-triangle property implies that there exists $R'>0$ such that, if

$$gQ\cap(\partial Y-O_R(o_Y,\xi_t))\neq\emptyset,$$

then

$$d(\xi_t,g\xi_t)<R'.$$

For such $g\in\mathsf{P}$, we therefore have

$$\nu_{\xi_t}(gQ)\asymp\nu_{\xi_t}(Q).$$

After increasing R by a uniform amount, we may also assume that

$$Q\subset O_R(\xi_t,o_Y).$$

Hence Lemma 6.5 gives

$$\nu_{\xi_t}(Q) \asymp e^{-\mathsf{d}_\psi(\xi_t, o_Y)} \nu(Q).$$

Using Proposition 5.2, we obtain

$$\nu_{\xi_t}(\partial Y - O_R(o_Y, \xi_t)) \asymp e^{-\mathsf{d}_\psi(\xi_t, o_Y)} e^{2\delta_{\bar\psi}(\mathsf{P})\mathsf{d}_{\bar\psi}(o_Y, \xi_t)} \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(o_Y, \xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}$$

as claimed. □

7. General form of the global shadow lemma

Continuing with the setting of section 6, we now prove the general form of the global shadow lemma. The estimate below should be viewed as the cusp version of the usual shadow lemma: when the point under consideration stays outside the horoballs in $\mathcal{B}$, one recovers the ordinary orbit-shadow estimate (Theorem 1.3).

Recall that, for $\xi \in \partial Y$ and $t \geq 0$, we write $\xi_t \in [o_Y, \xi]$ for the point satisfying $d(o_Y, \xi_t) = t$.

Theorem 7.1 (Global Shadow Lemma)**.** *For all sufficiently large $R > 0$, the following estimate holds uniformly. Let $\xi \in \partial Y$ and suppose that $\xi_t \in B_\eta$ for some $t \geq 0$ and some horoball $B_\eta \in \mathcal{B}$ based at $\eta \in \Gamma\xi_{\mathsf{P}}$, where $\mathsf{P} \in \mathcal{P}$. Let $\gamma \in \Gamma$ be such that γo_Y is a closest orbit point to ξ_t in the orbit Γo_Y, with respect to the metric d on Y. Then*

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{-\mathsf{d}_\psi(o_Y, \xi_t)} e^{\mathsf{d}_\psi(\gamma o_Y, \xi_t)} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma o_Y, \xi_t)} \cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\gamma o_Y, \xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.$$

Equivalently,

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{-\mathsf{d}_\psi(o_Y, \xi_t)} e^{\mathsf{d}_\psi(\Gamma o_Y, \xi_t)} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\Gamma o_Y, \xi_t)} \cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\Gamma o_Y, \xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.$$

The implied constants are independent of ξ, t, η, and γ.

Proof. Let $A > 0$ be a sufficiently large constant depending only on the hyperbolicity constant of Y and on the uniform constants fixed above. We allow A to increase finitely many times during the proof.

First note that the endpoint of

$$[o_Y, \xi] \cap \partial B_\eta$$

closest to ξ_t is uniformly close to some orbit point of Γo_Y. Consequently, if γo_Y is chosen to be a closest orbit point to ξ_t, for some $\gamma \in \Gamma$, then Lemma 4.11 gives

$$\mathsf{d}_\psi(\gamma o_Y, \xi_t) \approx \mathsf{d}_\psi(\Gamma o_Y, \xi_t) \quad \text{and} \quad \mathsf{d}_{\bar\psi}(\gamma o_Y, \xi_t) \approx \mathsf{d}_{\bar\psi}(\Gamma o_Y, \xi_t).$$

Thus it suffices to prove the estimate with this choice of γ.

We divide the proof into three cases.

Case 1: $\eta \in O_{R-A}(o_Y, \xi_t)$**.** In this case, there exists $x \in [o_Y, \eta]$ such that

$$d(x, \xi_t) < R - A.$$

This implies

$$O_A(o_Y, x) \subset O_R(o_Y, \xi_t) \subset O_{2R-A}(o_Y, x).$$

Note that we may also choose $\gamma \in \Gamma$ so that γo_Y is uniformly close to $[o_Y, \eta]$, in this case. Applying Theorem 6.3 to the parabolic point η, and then using Lemma 2.1 to replace x by ξ_t, we obtain

$$\nu(O_R(o_Y, \xi_t)) \asymp e^{-\psi(\mu(\gamma))} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma o_Y, \xi_t)} \cdot \left(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\gamma o_Y, \xi_t)\right)^{a_{\bar\psi}(\mathsf{P})}.$$

Since ξ_t lies in B_η and is uniformly close to the ray $[o_Y, \eta]$, Lemmas 4.11 and 4.5 imply

$$-\psi(\mu(\gamma)) = -\mathsf{d}_\psi(o_Y, \gamma o_Y) \approx -\mathsf{d}_\psi(o_Y, \xi_t) + \mathsf{d}_\psi(\gamma o_Y, \xi_t).$$

Substituting this into the preceding estimate proves the theorem in this case.

Case 2: $\eta \notin O_{R+A}(o_Y, \xi_t)$**.** In particular, we have $\xi \neq \eta$. Let $o' \in [o_Y, \xi] \cap \partial B_\eta$ be the endpoint farthest from o_Y; equivalently, o' is the point at which the ray $[o_Y, \xi]$ exits the horoball B_η. Set

$$t' := d(o', \xi_t).$$

We first compare the shadow $O_R(o_Y, \xi_t)$ with complements of shadows based at o'. If $\zeta \in O_R(o_Y, \xi_t)$, then $d([o_Y, \zeta], \xi_t) < R$, and hence

$$d(o', [o_Y, \zeta]) \le d(o', \xi_t) + R.$$

Therefore

$$O_R(o_Y, \xi_t) \subset \partial Y - \left\{\zeta \in \partial Y : d(o', [o_Y, \zeta]) > d(o', \xi_t) + R\right\}. \tag{7.1}$$

Conversely, if

$$d(o', [o_Y, \zeta]) \le d(o', \xi_t) + R - O(\delta),$$

then by the Gromov hyperbolicity, this implies that $[o_Y, \zeta]$ passes within distance R of ξ_t, provided $O(\delta)$ is sufficiently large. Hence

$$\partial Y - \left\{\zeta \in \partial Y : d(o', [o_Y, \zeta]) > d(o', \xi_t) + R - O(\delta)\right\} \subset O_R(o_Y, \xi_t). \tag{7.2}$$

For $s \ge 0$, let $\eta_s \in [o', \eta]$ be the point satisfying $d(o', \eta_s) = s$. We claim that

$$O_R(o_Y, \xi_t) \subset \partial Y - O_{A/2}(o', \eta_{t'+R+A}) \tag{7.3}$$

and

$$\partial Y - O_{A/2}(o', \eta_{t'+R}) \subset O_R(o_Y, \xi_t). \tag{7.4}$$

To prove (7.3), let

$$\zeta \in O_{A/2}(o', \eta_{t'+R+A}).$$

By Lemma 3.3,

$$\langle \zeta, \eta \rangle_{o'} \ge t' + R + A/2 - O(\delta).$$

Since $\eta \notin O_{R+A}(o_Y, \xi_t)$, applying (7.2) with $R + A$ in place of R gives

$$d(o', [o_Y, \eta]) > t' + R + A - O(\delta).$$

The Gromov product inequality (3.1) then implies

$$\langle o_Y, \zeta\rangle_{o'} \geq t' + R + A/2 - O(\delta).$$

For A sufficiently large, this and (7.1) imply that $\zeta \notin O_R(o_Y, \xi_t)$. This proves (7.3).

The proof of (7.4) is similar. If $\zeta \notin O_R(o_Y, \xi_t)$, then by (7.2),

$$d(o', [o_Y, \zeta]) > t' + R - O(\delta).$$

Together with the estimate for $d(o', [o_Y, \eta])$ above, the Gromov product inequality gives

$$\langle \zeta, \eta\rangle_{o'} \geq t' + R - O(\delta).$$

By Lemma 3.3, this implies

$$\zeta \in O_{A/2}(o', \eta_{t'+R})$$

after increasing A, if necessary. This proves (7.4).

We now estimate the measure. The hypothesis $\eta \notin O_{R+A}(o_Y, \xi_t)$ implies, by thin triangles, that $\eta_{t'}$ and ξ_t are uniformly close. Using (7.3), moving the basepoint from $\eta_{t'}$ to $\eta_{t'+R+A}$, and absorbing the resulting multiplicative constant into $\asymp$, we get

$$\begin{aligned}\nu_{\xi_t}(O_R(o_Y, \xi_t)) &\ll \nu_{\eta_{t'}}\big(\partial Y - O_{A/2}(o', \eta_{t'+R+A})\big)\\ &\ll \nu_{\eta_{t'+R+A}}\big(\partial Y - O_{A/2}(o', \eta_{t'+R+A})\big).\end{aligned}$$

Since $o' \in \partial B_\eta$, there exists $\gamma_0 \in \Gamma$ such that $\eta = \gamma_0 \xi_{\mathsf{P}}$ and $d(o', \gamma_0 o_Y)$ is uniformly bounded. By equivariance of the measures ν_x,

$$\begin{aligned}&\nu_{\eta_{t'+R+A}}\big(\partial Y - O_{A/2}(o', \eta_{t'+R+A})\big)\\ &\quad\asymp \nu_{\gamma_0^{-1}\eta_{t'+R+A}}\big(\partial Y - O_{A/2}(o_Y, \gamma_0^{-1}\eta_{t'+R+A})\big).\end{aligned}$$

Applying Proposition 6.8, we obtain

$$\begin{aligned}&\nu_{\gamma_0^{-1}\eta_{t'+R+A}}\big(\partial Y - O_{A/2}(o_Y, \gamma_0^{-1}\eta_{t'+R+A})\big)\\ &\quad\asymp e^{-\mathsf{d}_\psi(\gamma_0^{-1}\eta_{t'+R+A}, o_Y)} e^{2\delta_{\bar\psi}(\mathsf{P})\mathsf{d}_{\bar\psi}(o_Y, \gamma_0^{-1}\eta_{t'+R+A})}\\ &\qquad\cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(o_Y, \gamma_0^{-1}\eta_{t'+R+A})\big)^{a_{\bar\psi}(\mathsf{P})}\\ &\quad\asymp e^{\mathsf{d}_\psi(\gamma_0 o_Y, \eta_{t'+R+A})} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma_0 o_Y, \eta_{t'+R+A})}\\ &\qquad\cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\gamma_0 o_Y, \eta_{t'+R+A})\big)^{a_{\bar\psi}(\mathsf{P})}.\end{aligned}$$

Combining altogether gives

$$\begin{aligned}\nu_{\xi_t}(O_R(o_Y, \xi_t)) \ll\ & e^{\mathsf{d}_\psi(\gamma_0 o_Y, \eta_{t'+R+A})} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma_0 o_Y, \eta_{t'+R+A})}\\ &\cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\gamma_0 o_Y, \eta_{t'+R+A})\big)^{a_{\bar\psi}(\mathsf{P})}.\end{aligned}$$

The points $\eta_{t'+R+A}$ and ξ_t are at uniformly bounded distance depending only on R and A. Therefore, by Lemmas 2.1 and 4.11,

$$\mathsf{d}_\psi(\gamma_0 o_Y, \eta_{t'+R+A}) \approx \mathsf{d}_\psi(\gamma o_Y, \xi_t),$$

and similarly for $\mathsf{d}_{\bar\psi}$. Hence

$$\nu_{\xi_t}(O_R(o_Y,\xi_t)) \ll e^{\mathsf{d}_\psi(\gamma o_Y,\xi_t)} e^{2(\delta_{\bar\psi}(\mathsf{P})-1)\mathsf{d}_{\bar\psi}(\gamma o_Y,\xi_t)} \cdot \big(C_{\bar\psi} + \mathsf{d}_{\bar\psi}(\gamma o_Y,\xi_t)\big)^{a_{\bar\psi}(\mathsf{P})}.$$

The reverse inequality is obtained in the same way using (7.4) instead of (7.3).

Finally, Lemma 6.4 gives

$$\nu(O_R(o_Y,\xi_t)) \asymp e^{-\mathsf{d}_\psi(o_Y,\xi_t)} \nu_{\xi_t}(O_R(o_Y,\xi_t)),$$

and the desired estimate follows.

Case 3: $\eta \notin O_{R-A}(o_Y,\xi_t)$ and $\eta \in O_{R+A}(o_Y,\xi_t)$. Take R sufficiently large that $R-2A$ exceeds all the preceding thresholds. Since $\eta \notin O_{R-A}(o_Y,\xi_t)$, Case 2 applies to $O_{R-2A}(o_Y,\xi_t)$. Since $\eta \in O_{R+A}(o_Y,\xi_t)$, Case 1 applies to $O_{R+2A}(o_Y,\xi_t)$. Both estimates have the same right-hand side, up to uniform multiplicative constants. The inclusions

$$O_{R-2A}(o_Y,\xi_t) \subset O_R(o_Y,\xi_t) \subset O_{R+2A}(o_Y,\xi_t)$$

therefore give the desired estimate.

The proof is complete. □

Remark 7.2. When $\operatorname{rank} G = 1$, the relatively Morse condition coincides with geometric finiteness. Moreover, after choosing the unit vector $H_0 \in \mathfrak{a}^+$, we identify $\mathfrak{a}$ with $\mathbb{R}H_0$ and take $\psi(tH_0) = \delta_\Gamma t$. Then $\psi(\mu(g)) = \delta_\Gamma d(o, go)$ and $\delta_\psi(\Gamma) = 1$. Since the opposition involution is trivial in rank one, $\bar\psi = \psi$; and by Theorem 5.5, we have $a_\psi(\mathsf{P}) = a_{\bar\psi}(\mathsf{P}) = 0$. When $G = \mathrm{SO}(n,1)^\circ$, Theorem 7.1 therefore specializes to Theorem 1.2. In this sense, Theorem 7.1 is the higher-rank relatively Morse analogue of the global shadow lemma of Stratmann–Velani [24]. The condition $\delta_\psi(\Gamma) = 1$ in Theorem 7.1 is merely a normalization of the Patterson–Sullivan parameter; in the rank-one specialization above, it is achieved by the choice $\psi(tH_0) = \delta_\Gamma t$.

8. Local properties of Patterson–Sullivan measures

In this section, we apply the global shadow lemma to local properties of Patterson–Sullivan measures. As before, let $\Gamma < G$ be θ-Morse relative to $\mathcal{P}$, with Morse embedding $f : Y \to X$ of a Gromov model (Y,d) for $(\Gamma,\mathcal{P})$. Let $\psi \in \mathfrak{a}_\theta^*$ be such that

$$\psi > 0 \quad \text{on } \mathcal{L}_f - \{0\} \quad \text{and} \quad \delta_\psi(\Gamma) = 1.$$

Let ν be a (Γ,ψ)-Patterson–Sullivan measure on Λ_θ. Throughout this section, we also assume that $\theta = \mathrm{i}(\theta)$.

The visual quasi-metric. We first define the higher-rank visual quasi-metric associated to ψ. For distinct $\xi, \eta \in \Lambda_\theta$, set

$$d_\psi(\xi, \eta) := e^{-\psi(\mathcal{G}^\theta(\xi,\eta))}, \tag{8.1}$$

and put $d_\psi(\xi, \xi) = 0$. Note that this is not the same object as the metric-like function d_ψ on Y. This function behaves like a metric: there exists $c > 0$ such that

$$d_\psi(\xi, \eta) \le c\big(d_\psi(\xi, \zeta) + d_\psi(\zeta, \eta)\big) \tag{8.2}$$

for all $\xi, \eta, \zeta \in \Lambda_\theta$. For Anosov groups, this was proved in [19, Proposition 5.3]; the same argument applies to a general Morse embedding.

For $r > 0$ and $\xi \in \Lambda_\theta$, let

$$B_\psi(\xi, r) := \{\eta \in \Lambda_\theta : d_\psi(\xi, \eta) < r\}.$$

By Lemma 4.4, d_ψ and $d_{\bar\psi}$ are bi-Lipschitz equivalent, since $\frac{1}{2}(\mu_\theta(f(z)) + \mathrm{i}\,\mu_\theta(f(z)))$ is i-invariant and hence ψ and $\bar\psi$ agree on it. In particular, B_ψ-balls and $B_{\bar\psi}$-balls are comparable.

We identify ∂Y with Λ_θ via the Γ-equivariant homeomorphism $f : \partial Y \to \Lambda_\theta$. Proposition 4.12 implies that, for all sufficiently large $R > 0$, there exist constants $c_1, c_2 > 0$ such that

$$B_\psi\big(\xi, c_1 e^{-\mathsf{d}_{\bar\psi}(o_Y, \xi_t)}\big) \subset O_R(o_Y, \xi_t) \subset B_\psi\big(\xi, c_2 e^{-\mathsf{d}_{\bar\psi}(o_Y, \xi_t)}\big) \tag{8.3}$$

for all $\xi \in \Lambda_\theta$ and all $t \ge 0$, where $\xi_t \in [o_Y, \xi]$ is the point satisfying $d(o_Y, \xi_t) = t$.

For later use, we rewrite the global shadow lemma in a compact form. If $x \in Y$ lies in a horoball based at a point of $\Gamma\xi_{\mathsf{P}}$, choose $\gamma_x \in \Gamma$ so that $\gamma_x o_Y$ is a closest orbit point to x, and set

$$h_\psi(x) := \mathsf{d}_\psi(\gamma_x o_Y, x), \quad \delta(x) := \delta_\psi(\mathsf{P}), \quad \text{and} \quad a(x) := a_{\bar\psi}(\mathsf{P}).$$

If x lies outside the horoballs, we set

$$h_\psi(x) = h_{\bar\psi}(x) = 0, \quad \delta(x) = 0, \quad \text{and} \quad a(x) = 0.$$

By Proposition 4.3, $\psi(\mu_\theta(\gamma)) = \bar\psi(\mu_\theta(\gamma)) + O(1)$ uniformly for $\gamma \in \mathsf{P}$. Hence the corresponding Poincaré series have the same abscissa of convergence, so $\delta_\psi(\mathsf{P}) = \delta_{\bar\psi}(\mathsf{P})$ for every $\mathsf{P} \in \mathcal{P}$. Define

$$F(x) := \mathsf{d}_\psi(o_Y, x) - h_\psi(x) + 2(1 - \delta(x))h_{\bar\psi}(x).$$

The global shadow lemma (Theorem 7.1) then takes the form

$$\nu(O_R(o_Y, x)) \asymp e^{-F(x)}(C_{\bar\psi} + h_{\bar\psi}(x))^{a(x)}, \tag{8.4}$$

with implied constants independent of x.

Lemma 8.1. *For every sufficiently large $R > 0$, there exists $C = C(R) > 1$ such that, for every $y \in Y$ and every $x \in [o_Y, y]$,*

$$C^{-1}d(x, y) - C \le \log \frac{\nu(O_R(o_Y, x))}{\nu(O_R(o_Y, y))} \le Cd(x, y) + C.$$

Proof. By tautness and hyperbolicity, there exists a uniform $D > 0$ such that x, y are within distance D of ordered points x', y' on a geodesic ray from o_Y. Taking R sufficiently large, the inclusions

$$O_{R-D}(o_Y, z') \subset O_R(o_Y, z) \subset O_{R+D}(o_Y, z')$$

and the global shadow lemma give $\nu(O_R(o_Y, z)) \asymp \nu(O_R(o_Y, z'))$ for $z = x, y$. Since $|d(x, y) - d(x', y')| \le 2D$, it suffices to consider x, y on a common geodesic ray from o_Y.

First suppose that both x and y are uniformly close to Γo_Y. Then h_ψ and $h_{\bar\psi}$ are uniformly bounded at both points. By (8.4) and Lemma 4.5,

$$\log \frac{\nu(O_R(o_Y, x))}{\nu(O_R(o_Y, y))} \approx \mathsf{d}_\psi(o_Y, y) - \mathsf{d}_\psi(o_Y, x) \approx \mathsf{d}_\psi(x, y).$$

The desired bounds follow from Lemma 4.10.

Now suppose that x and y belong to the same horoball based at p. If the geodesic ray from o_Y containing x, y exits the horoball, let u, v be its entrance and exit points, and let m be the midpoint of $[u, v]$. Since u, v lie on the boundary of the same horoball, hyperbolicity implies that $[u, m]$ and $[m, v]$ stay uniformly close to $[u, p]$ and $[v, p]$, respectively. Applying Lemma 4.11 to $\psi \circ \mathsf{i}$, after translating orbit points uniformly close to u, v, and using Lemma 4.5, we obtain:

(1) If y lies in the first half of $[u, v]$, then

$$h_\psi(y) - h_\psi(x) \approx \mathsf{d}_\psi(x, y).$$

(2) If x lies in the second half of $[u, v]$, then

$$h_\psi(y) - h_\psi(x) \approx -\mathsf{d}_\psi(y, x).$$

Here we used $\mathsf{d}_{\psi\circ\mathsf{i}}(z, w) = \mathsf{d}_\psi(w, z)$. The same statements hold with ψ replaced by $\bar\psi$. If the ray does not exit the horoball, it ends at p, and the same argument gives (1) throughout its terminal part in the horoball.

Write $\delta = \delta(x) = \delta(y)$ and $a = a(x) = a(y)$. By Theorem 5.4 and the finiteness of $\mathcal{P}$, both δ and $1 - \delta$ are uniformly bounded away from zero, and a is uniformly bounded.

In case (1), Lemma 4.5 gives

$$\begin{aligned} F(y) - F(x) &\approx \mathsf{d}_\psi(x, y) - \big(h_\psi(y) - h_\psi(x)\big) + 2(1-\delta)\big(h_{\bar\psi}(y) - h_{\bar\psi}(x)\big) \\ &\approx 2(1-\delta)\mathsf{d}_{\bar\psi}(x, y). \end{aligned}$$

We also have

$$\begin{aligned} \left|\log \frac{C_{\bar\psi} + h_{\bar\psi}(x)}{C_{\bar\psi} + h_{\bar\psi}(y)}\right| &\ll 1 + \log\big(1 + |h_{\bar\psi}(x) - h_{\bar\psi}(y)|\big) \\ &\ll 1 + \log(1 + d(x, y)). \end{aligned}$$

Since $\mathsf{d}_{\bar\psi}$ satisfies the quasi-isometric bounds in Lemma 4.10, the logarithmic correction can be absorbed into the positive linear term, using $\log(1 + t) \le \varepsilon t + C_\varepsilon$ with $\varepsilon > 0$ sufficiently small. The claim therefore follows from (8.4).

In case (2), we similarly have

$$F(y) - F(x) \approx 2\delta\, \mathsf{d}_{\bar\psi}(x, y),$$

using $\mathsf{d}_\psi(x, y) + \mathsf{d}_\psi(y, x) = 2\mathsf{d}_{\bar\psi}(x, y)$. Together with the preceding bound on the logarithmic correction, the same argument proves the claim in this case.

If x, y lie in the same horoball and $m \in [x, y]$, apply the bounds proved in case (1) to (x, m) and those proved in case (2) to (m, y). Adding them gives the desired bounds for (x, y).

Finally, suppose that x, y lie in different horoballs, allowing either point to lie outside the horoballs. Let x_1 be the exit point from the horoball containing x, taking $x_1 = x$ if x lies outside the horoballs. Similarly, let y_1 be the entrance point into the horoball containing y, taking $y_1 = y$ if y lies outside the horoballs. Then $[x, y] = [x, x_1] \cup [x_1, y_1] \cup [y_1, y]$. Each of the initial and terminal subsegments splits into at most two pieces covered by (1) or (2). Both endpoints of the middle subsegment are uniformly close to Γo_Y, so the first case of the proof applies to it, regardless of the horoballs it crosses. Thus there are at most 5 pieces. Summing their estimates proves the lemma. □

Remark 8.2. When $\psi = \psi \circ \mathrm{i}$, using (8.3), (8.4) can also be written in terms of d_ψ-balls: for $\xi \in \Lambda_\theta$ and $r > 0$, let $x \in [o_Y, \xi]$ be such that $\mathsf{d}_\psi(o_Y, x) \approx -\log r$ with a uniform additive error. Then

$$\nu(B_\psi(\xi, r)) \asymp r e^{(2\delta(x)-1)h_\psi(x)} (C_\psi + h_\psi(x))^{a(x)}.$$

Local doubling and reverse doubling. We now deduce that ν is locally doubling and locally reverse doubling with respect to d_ψ. The latter means that shrinking the radius by a sufficiently large factor reduces the mass by any prescribed factor, uniformly in the center and the scale.

Theorem 8.3. (1) *For every $L > 1$, there exists $\varepsilon_L > 0$ such that*

$$\nu(B_\psi(\xi, r)) \geq \varepsilon_L\, \nu(B_\psi(\xi, Lr))$$

for all $\xi \in \Lambda_\theta$ and all $r > 0$ with $Lr \leq 1$.

(2) *For every $\kappa \geq 1$, there exists $L > 1$ such that*

$$\nu(B_\psi(\xi, r/L)) \leq \kappa^{-1} \nu(B_\psi(\xi, r))$$

for all $\xi \in \Lambda_\theta$ and all $0 < r \leq 1$.

Proof. Coarse additivity and Lemma 4.10 compare increments of $\mathsf{d}_{\bar\psi}(o_Y, \xi_t)$ with distances along the ray, up to uniform multiplicative and additive constants. Combining this with Lemma 8.1 and (8.3), we obtain constants $C, c > 1$ such that

$$C^{-1} \left(\frac{r}{s}\right)^c \leq \frac{\nu(B_\psi(\xi, r))}{\nu(B_\psi(\xi, s))} \leq C \left(\frac{r}{s}\right)^{1/c}$$

for every $\xi \in \Lambda_\theta$ and $0 < r \leq s \leq 1$.

Taking $s = Lr$ proves (1) with $\varepsilon_L = C^{-1}L^{-c}$. For (2), choose $L \geq (C\kappa)^c$ and apply the upper bound to the radii r/L and r. □

9. Hausdorff measure

In this section, we characterize when the Patterson–Sullivan measure agrees, up to scale, with the Hausdorff measure defined by the visual quasi-metric. As in the preceding sections, let $\Gamma < G$ be θ-Morse relative to $\mathcal{P}$, with Morse embedding

$$f : Y \to X$$

from a Gromov model (Y, d) for $(\Gamma, \mathcal{P})$. Let $\psi \in \mathfrak{a}_\theta^*$ satisfy

$$\psi > 0 \quad \text{on } \mathcal{L}_f - \{0\}, \qquad \delta_\psi(\Gamma) = 1.$$

Let ν be the corresponding Patterson–Sullivan probability measure on Λ_θ. We assume that $\theta = \mathrm{i}(\theta)$ and write

$$\bar{\psi} := \frac{\psi + \psi \circ \mathrm{i}}{2}.$$

We identify ∂Y with Λ_θ via f.

Equip Λ_θ with the visual quasi-metric d_ψ defined in (8.1). The quasi-triangle inequality in (8.2), together with $d_\psi \asymp d_{\bar{\psi}}$ and the symmetry of $d_{\bar{\psi}}$, gives the usual covering lemma: from a family of balls with uniformly bounded radii, one can select a countable disjoint subfamily whose expansions by a uniform factor cover the centers of the original balls; see, for instance, [19, Lemma 6.12]. We use this form of the Vitali covering lemma below.

For $s > 0$, $\varepsilon > 0$, and $E \subset \Lambda_\theta$, define

$$\mathcal{H}^s_{\psi,\varepsilon}(E) := \inf \left\{ \sum_i (\operatorname{diam}_\psi U_i)^s : E \subset \bigcup_i U_i, \quad \sup_i \operatorname{diam}_\psi U_i \leq \varepsilon \right\},$$

where $\operatorname{diam}_\psi U := \sup_{\xi,\eta \in U} d_\psi(\xi, \eta)$. The outer measure

$$\mathcal{H}^s_\psi(E) := \lim_{\varepsilon \to 0} \mathcal{H}^s_{\psi,\varepsilon}(E)$$

restricts to a Borel measure on Λ_θ; see [13] and [11, Appendix A]. We call it the s-dimensional Hausdorff measure associated to d_ψ. For $s = 1$, write $\mathcal{H}_{\psi,\varepsilon} := \mathcal{H}^1_{\psi,\varepsilon}$ and $\mathcal{H}_\psi := \mathcal{H}^1_\psi$.

Theorem 9.1. *If Γ is Zariski dense, then ν is a positive multiple of $\mathcal{H}_\psi$ if and only if both of the following hold:*

(1) $\psi = \psi \circ \mathrm{i}$*;*
(2) *for every $\mathsf{P} \in \mathcal{P}$,*

$$\delta_\psi(\mathsf{P}) < \frac{1}{2}, \quad \text{or} \quad \delta_\psi(\mathsf{P}) = \frac{1}{2} \text{ and } a_\psi(\mathsf{P}) = 0.$$

For arbitrary Γ, if ψ is symmetric, then ν is a positive multiple of $\mathcal{H}_\psi$ if and only if condition (2) holds.

Remark 9.2. This theorem generalizes Sullivan's Hausdorff-measure criterion for geometrically finite Kleinian groups [26]. In the real hyperbolic case, a rank-k parabolic subgroup has critical exponent $\delta_{\mathsf{P}} = k/2$. After the normalization $\delta_\psi(\Gamma) = 1$, we have $\delta_\psi(\mathsf{P}) = \frac{\delta_{\mathsf{P}}}{\delta_\Gamma}$ and $a_\psi(\mathsf{P}) = 0$. Thus the condition in the theorem becomes $k \le \delta_\Gamma$ for every cusp. Anosov groups are the case with no peripheral subgroups, so the theorem also generalizes [12, Theorem 1.1].

If ψ is symmetric and $0 < \mathcal{H}_\psi(\Lambda_\theta) < \infty$, the Hausdorff measure has the same conformality rule as ν:

$$\frac{d\gamma_*\mathcal{H}_\psi}{d\mathcal{H}_\psi}(\xi) = e^{\psi(\beta_\xi^\theta(e,\gamma))} \qquad (\gamma \in \Gamma).$$

The proof of [12, Lemma 9.7], stated there for Anosov groups, applies unchanged, since it uses the transformation rule for d_ψ. Uniqueness of the Patterson–Sullivan probability measure (Theorem 6.1) then gives

$$\mathcal{H}_\psi = \mathcal{H}_\psi(\Lambda_\theta)\,\nu. \tag{9.1}$$

We use this identity only for symmetric ψ with positive and finite total Hausdorff measure.

Local estimates. Denote by $\Lambda_\theta^{\mathrm{con}}$ the image of the conical limit set in ∂Y. We first establish the local estimates for symmetric ψ. The general finiteness statement will then follow by applying these estimates to a normalized multiple of $\bar\psi$.

Lemma 9.3. *Assume that $\psi = \psi \circ \mathrm{i}$. There exists $C > 1$ such that:*

(1) *For every $\xi \in \Lambda_\theta^{\mathrm{con}}$, there exists a sequence $r_i \to 0$ such that*
$$\nu(B_\psi(\xi, r_i)) \ge C^{-1} r_i \qquad \text{for all } i.$$

(2) *If the peripheral condition in Theorem 9.1 holds, then, after increasing C if necessary,*
$$\nu(B_\psi(\xi, r)) \le Cr \qquad (\xi \in \Lambda_\theta,\ r > 0).$$

Proof. Fix $R > 0$ sufficiently large for (8.3) and (8.4) to hold. Since $\psi = \bar\psi$, the latter estimate becomes

$$\nu(O_R(o_Y, x)) \asymp e^{-\mathsf{d}_\psi(o_Y,x)} e^{(2\delta(x)-1)h_\psi(x)} (C_\psi + h_\psi(x))^{a(x)}. \tag{9.2}$$

All implied constants are independent of x.

For (1), fix $\xi \in \Lambda_\theta^{\mathrm{con}}$. The ray $[o_Y, \xi]$ meets $Y - \bigcup_{B\in\mathcal{B}} B$ at arbitrarily large times. Indeed, otherwise a tail of the ray would lie in a single horoball, so ξ would be its parabolic basepoint, contrary to conicality. Choose $x_i \in [o_Y, \xi] - \bigcup_{B\in\mathcal{B}} B$ with $d(o_Y, x_i) \to \infty$. These points are uniformly close to Γo_Y by cocompactness of the thick part, and $h_\psi(x_i) = 0$. Hence (9.2) gives

$$\nu(O_R(o_Y, x_i)) \asymp e^{-\mathsf{d}_\psi(o_Y,x_i)}$$

with constants independent of ξ and i. By (8.3), there is a uniform $c_2 > 0$ such that, for $r_i := c_2 e^{-\mathsf{d}_\psi(o_Y, x_i)}$,

$$O_R(o_Y, x_i) \subset B_\psi(\xi, r_i).$$

Lemma 4.10 gives $r_i \to 0$, proving (1).

To prove (2), note that by the hypothesis on the parabolic subgroups, the cusp correction factor in the global shadow lemma is uniformly bounded above. Indeed, if x lies in a horoball associated to P, then the correction factor is

$$e^{(2\delta_\psi(\mathsf{P})-1)h_\psi(x)}\big(C_\psi + h_\psi(x)\big)^{a_\psi(\mathsf{P})}.$$

This is uniformly bounded when $\delta_\psi(\mathsf{P}) < 1/2$, and also when $\delta_\psi(\mathsf{P}) = 1/2$ and $a_\psi(\mathsf{P}) = 0$. Here we use that h_ψ is uniformly bounded below and $\mathcal{P}$ is finite. In the thick part, the usual shadow estimate gives the same conclusion. Hence

$$\nu(O_R(o_Y, x)) \ll e^{-\mathsf{d}_\psi(o_Y, x)} \quad \text{uniformly for all } x \in Y.$$

Now fix $\xi \in \Lambda_\theta$ and let $r > 0$ be sufficiently small. Let c_1 be the constant in (8.3), and let A be the constant in Lemma 4.10(3). Apply that lemma with $s = \log(c_1/r) - A$ to choose $x \in [o_Y, \xi]$ such that

$$\log(c_1/r) - 2A \le \mathsf{d}_\psi(o_Y, x) \le \log(c_1/r).$$

Thus $r \le c_1 e^{-\mathsf{d}_\psi(o_Y, x)} \ll r$, and (8.3) gives $B_\psi(\xi, r) \subset O_R(o_Y, x)$. Then

$$\nu(B_\psi(\xi, r)) \le \nu(O_R(o_Y, x)) \ll e^{-\mathsf{d}_\psi(o_Y, x)} \asymp r.$$

After increasing the constant, the same bound holds for all $r > 0$, since ν is a probability measure. □

Finiteness and positivity.

Proposition 9.4. *The total Hausdorff measure satisfies*

$$\mathcal{H}_\psi(\Lambda_\theta) < \infty.$$

It is positive if and only if $\delta_{\bar\psi}(\Gamma) = 1$ *and, for every* $\mathsf{P} \in \mathcal{P}$,

$$\delta_{\bar\psi}(\mathsf{P}) < \frac{1}{2}, \quad \textit{or} \quad \delta_{\bar\psi}(\mathsf{P}) = \frac{1}{2} \textit{ and } a_{\bar\psi}(\mathsf{P}) = 0.$$

If Γ *is Zariski dense, these conditions are equivalent to conditions (1) and (2) of Theorem 9.1. In particular, for Zariski dense* Γ *and nonsymmetric* ψ, $\mathcal{H}_\psi(\Lambda_\theta) = 0$.

Proof. We first prove the proposition when ψ is symmetric. Every limit point is either conical or parabolic. The parabolic limit set is countable and therefore has $\mathcal{H}_\psi$-measure zero. It therefore suffices to show that $\mathcal{H}_\psi(\Lambda_\theta^{\mathrm{con}}) < \infty$. Fix $\varepsilon > 0$. For each $\xi \in \Lambda_\theta^{\mathrm{con}}$, choose $0 < r_\xi < \varepsilon$ such that $\nu(B_\psi(\xi, r_\xi)) \ge C^{-1} r_\xi$, using Lemma 9.3(1).

Applying the Vitali covering lemma to the family $\{B_\psi(\xi, r_\xi) : \xi \in \Lambda_\theta^{\mathrm{con}}\}$, there exists a countable disjoint subcollection $\{B_\psi(\xi_n, r_n) : n \in \mathbb{N}\}$ such that $\Lambda_\theta^{\mathrm{con}} \subset \bigcup_n B_\psi(\xi_n, \lambda r_n)$ for some uniform constant $\lambda > 1$. Since d_ψ

satisfies the triangle inequality up to a multiplicative constant (8.2), there exists $D > 0$ such that

$$\operatorname{diam}_\psi B_\psi(\xi_n, \lambda r_n) \le D r_n \quad \text{for all } n.$$

Hence

$$\mathcal{H}_{\psi, D\varepsilon}(\Lambda_\theta^{\mathrm{con}}) \le \sum_n D r_n \le DC \sum_n \nu(B_\psi(\xi_n, r_n)) \le DC\, \nu(\Lambda_\theta).$$

Since $\varepsilon > 0$ is arbitrary, this proves $\mathcal{H}_\psi(\Lambda_\theta^{\mathrm{con}}) < \infty$.

Suppose next that the peripheral condition holds. Let $\{U_i\}$ be a countable cover of Λ_θ with $\operatorname{diam}_\psi U_i \le \varepsilon$. For each nonempty U_i, choose $\xi_i \in U_i$ and $\rho_i > \operatorname{diam}_\psi U_i$ so that

$$\sum_i \rho_i \le \varepsilon + \sum_i \operatorname{diam}_\psi U_i.$$

Since $U_i \subset B_\psi(\xi_i, \rho_i)$, Lemma 9.3(2) gives

$$1 \le \sum_i \nu(B_\psi(\xi_i, \rho_i)) \le C \sum_i \rho_i \le C \left(\varepsilon + \sum_i \operatorname{diam}_\psi U_i \right).$$

Taking the infimum and then letting $\varepsilon \to 0$ proves $\mathcal{H}_\psi(\Lambda_\theta) > 0$.

It remains to prove that a peripheral subgroup violating the condition in Theorem 9.1 forces the Hausdorff measure to vanish. We first show that almost every ray enters translates of any fixed peripheral horoball to arbitrarily large depth.

Fix $\mathsf{P} \in \mathcal{P}$, let U_{P} be the union of all Γ-translates of its horoball, and set

$$b_{\mathsf{P}}(z) := d(z, Y - U_{\mathsf{P}}) \qquad (z \in Y).$$

This is a Γ-invariant 1-Lipschitz function.

The diagonal Γ-action on $(\Lambda_\theta^{(2)}, \nu \otimes \nu)$ is ergodic by [9, Corollary 1.5]. It follows that almost every pair (ξ, η) has a dense Γ-orbit in $\Lambda_\theta^{(2)}$.

Let B_{P} be the fixed horoball based at $p = \xi_{\mathsf{P}}$. We claim that for each integer $N > 0$, there is a nonempty open set $V_N \subset \Lambda_\theta^{(2)}$ such that every geodesic joining a pair in V_N contains a point $z \in B_{\mathsf{P}}$ with $b_{\mathsf{P}}(z) > N$. To show this, choose $q \in \partial Y - \{p\}$ and a geodesic $[p, q]$. Let $D > 0$ be a uniform constant for thin ideal quadrilaterals. For each integer $N > 0$, choose $z \in [p, q]$ sufficiently far toward p that

$$d(z, Y - B_{\mathsf{P}}) > N + D.$$

Choose disjoint open neighborhoods V_p, V_q of p, q sufficiently small that the sides $[p, \xi]$ and $[q, \eta]$ stay farther than D from z whenever $\xi \in V_p - \{p\}$ and $\eta \in V_q - \{q\}$. Such neighborhoods exist because the Gromov products $\langle p, \xi \rangle_z$ and $\langle q, \eta \rangle_z$ tend to infinity as $\xi \to p$ and $\eta \to q$. Thinness of the ideal quadrilateral with vertices p, q, η, ξ then forces every geodesic $[\xi, \eta]$ to pass within distance D of z. Thus it contains a point w satisfying

$$b_{\mathsf{P}}(w) \ge d(w, Y - B_{\mathsf{P}}) \ge d(z, Y - B_{\mathsf{P}}) - D > N.$$

In the preceding argument, take $\xi \neq p$ and $\eta \neq q$ so that the four ideal vertices are distinct. Since ∂Y has no isolated points, $V_N := (V_p - \{p\}) \times (V_q - \{q\})$ is a nonempty open subset of $\Lambda_\theta^{(2)}$ with the required property.

For a pair (ξ, η) with dense Γ-orbit, choose $\gamma_N \in \Gamma$ such that $\gamma_N(\xi, \eta) \in V_N$. Translating back and using the Γ-invariance of b_{P} shows that every geodesic joining ξ and η contains a point z_N where $b_{\mathsf{P}}(z_N) > N$. Thus, for almost every pair, b_{P} is unbounded on every joining geodesic.

Let E_{P} be the set of endpoints ξ for which b_{P} is unbounded on $[o_Y, \xi]$. This property does not depend on the choice of the ray, since rays from o_Y to the same endpoint stay uniformly close. The set E_{P} is Borel: its complement is the countable union, over $k \in \mathbb{N}$, of the compact sets of endpoints admitting a ray from o_Y on which $b_{\mathsf{P}} \leq k$. Here compactness follows from properness of Y and continuity of the endpoint map.

If $\nu(E_{\mathsf{P}}^c) > 0$, then for distinct $\xi, \eta \in E_{\mathsf{P}}^c$, thinness of the ideal triangle with vertices o_Y, ξ, η places every joining geodesic in a uniform neighborhood of $[o_Y, \xi] \cup [o_Y, \eta]$. Hence b_{P} is bounded on every such geodesic. Since ν is atomless, the set of distinct pairs in $E_{\mathsf{P}}^c \times E_{\mathsf{P}}^c$ has product measure $\nu(E_{\mathsf{P}}^c)^2 > 0$. For every such pair, b_{P} is bounded on joining geodesics, whereas the preceding statement says that it is unbounded for almost every distinct pair. This contradiction shows that $\nu(E_{\mathsf{P}}^c) = 0$, and hence

$$\nu(E_{\mathsf{P}}) = 1.$$

Now suppose that P violates the peripheral condition in Theorem 9.1. Since $a_\psi(\mathsf{P})$ is a nonnegative integer, either $\delta_\psi(\mathsf{P}) > 1/2$, or $\delta_\psi(\mathsf{P}) = 1/2$ and $a_\psi(\mathsf{P}) > 0$. For $\xi \in E_{\mathsf{P}}$, choose $x_n \in [o_Y, \xi] \cap U_{\mathsf{P}}$ with $b_{\mathsf{P}}(x_n) \to \infty$. Since

$$d(x_n, \Gamma o_Y) \geq b_{\mathsf{P}}(x_n) - b_{\mathsf{P}}(o_Y),$$

the quasi-isometric lower bound in Lemma 4.10, applied to the closest orbit point used to define h_ψ, implies $h_\psi(x_n) \to \infty$. Set $r_n := c_2 e^{-\mathsf{d}_\psi(o_Y, x_n)}$, where c_2 is the uniform constant in (8.3). Then $r_n \to 0$, and (9.2) gives

$$\frac{\nu(B_\psi(\xi, r_n))}{r_n} \gg e^{(2\delta_\psi(\mathsf{P})-1)h_\psi(x_n)} (C_\psi + h_\psi(x_n))^{a_\psi(\mathsf{P})} \to \infty.$$

Hence the Borel set

$$E_\infty := \left\{ \xi \in \Lambda_\theta : \limsup_{r \to 0} \frac{\nu(B_\psi(\xi, r))}{r} = \infty \right\}$$

has full ν-measure. For any $M > 0$ and $\varepsilon > 0$, every point of E_∞ is the center of arbitrarily small balls with $r < \varepsilon$ and $\nu(B_\psi(\xi, r)) \geq Mr$. Applying the same Vitali argument as in the finiteness proof yields

$$\mathcal{H}_{\psi, D\varepsilon}(E_\infty) \leq D \sum_n r_n \leq \frac{D}{M} \sum_n \nu(B_\psi(\xi_n, r_n)) \leq \frac{D}{M},$$

where D is independent of M and ε. Letting $\varepsilon \to 0$ and then $M \to \infty$ gives $\mathcal{H}_\psi(E_\infty) = 0$. If $\mathcal{H}_\psi(\Lambda_\theta)$ were positive, it would be finite by the first part of

the proof, and (9.1) would imply

$$\mathcal{H}_\psi(E_\infty) = \mathcal{H}_\psi(\Lambda_\theta)\nu(E_\infty) = \mathcal{H}_\psi(\Lambda_\theta) > 0,$$

a contradiction. Therefore $\mathcal{H}_\psi(\Lambda_\theta) = 0$.

We now consider arbitrary ψ. Since $\mathcal{L}_f = \mathrm{i}(\mathcal{L}_f)$, $\bar{\psi}$ is positive on $\mathcal{L}_f - \{0\}$. The length bounds in Lemma 4.10, together with $\delta_\psi(\Gamma) = 1$, imply $\delta_0 := \delta_{\bar{\psi}}(\Gamma) > 0$. Moreover, $\delta_{\psi\circ\mathrm{i}}(\Gamma) = \delta_\psi(\Gamma) = 1$, so [9, Corollary 1.9] gives $\delta_0 \le 1$. The linear form $\delta_0\bar{\psi}$ is symmetric, is positive on $\mathcal{L}_f - \{0\}$, and satisfies $\delta_{\delta_0\bar{\psi}}(\Gamma) = 1$. The symmetric case just proved, applied to $\delta_0\bar{\psi}$ and its Patterson–Sullivan measure, gives $\mathcal{H}_{\delta_0\bar{\psi}}(\Lambda_\theta) < \infty$. By definition,

$$d_{\delta_0\bar{\psi}} = d_{\bar{\psi}}^{\delta_0}, \qquad \mathcal{H}_{\bar{\psi}}^{\delta_0} = \mathcal{H}_{\delta_0\bar{\psi}}.$$

If $\delta_0 < 1$, then for every $\varepsilon > 0$,

$$\mathcal{H}^1_{\bar{\psi},\varepsilon}(\Lambda_\theta) \le \varepsilon^{1-\delta_0}\mathcal{H}^{\delta_0}_{\bar{\psi},\varepsilon}(\Lambda_\theta) \le \varepsilon^{1-\delta_0}\mathcal{H}_{\delta_0\bar{\psi}}(\Lambda_\theta).$$

Letting $\varepsilon \to 0$ gives $\mathcal{H}_{\bar{\psi}}(\Lambda_\theta) = 0$. If $\delta_0 = 1$, the symmetric case applied to $\bar{\psi}$ gives finiteness and precisely the stated peripheral criterion for positivity. Since $d_\psi \asymp d_{\bar{\psi}}$ by Lemma 4.4, their one-dimensional Hausdorff measures are comparable. This proves the general assertion.

Finally, if Γ is Zariski dense and ψ is nonsymmetric, [9, Corollary 1.10], applied to ψ and $\psi\circ\mathrm{i}$, gives $\delta_0 < 1$. If ψ is symmetric, then $\delta_0 = \delta_\psi(\Gamma) = 1$ and $\bar{\psi} = \psi$. This proves the last assertions. □

Proof of Theorem 9.1. Suppose first that ψ is symmetric. By Proposition 9.4, condition (2) is equivalent to $0 < \mathcal{H}_\psi(\Lambda_\theta) < \infty$. Together with (9.1), this proves the required equivalence without a Zariski-density assumption.

Now suppose that Γ is Zariski dense and that $\psi \neq \psi\circ\mathrm{i}$. We show, more generally, that ν is not comparable to $\mathcal{H}^s_\psi$ for any $s > 0$. Suppose otherwise. By Lemma 4.4, $d_\psi \asymp d_{\bar{\psi}}$, and hence

$$\mathcal{H}^s_{\bar{\psi}} \asymp \mathcal{H}^s_\psi \asymp \nu, \qquad 0 < \mathcal{H}^s_{\bar{\psi}}(\Lambda_\theta) < \infty.$$

Since $d_{s\bar{\psi}} = d^s_{\bar{\psi}}$, the conformality argument of [12, Lemma 9.7], applied to the symmetric form $s\bar{\psi}$, shows that the normalization of $\mathcal{H}^s_{\bar{\psi}}$ is $(\Gamma, s\bar{\psi})$-conformal. This measure is absolutely continuous with respect to ν. Since ν is of divergence type, [16, Theorem 1.6] implies $\psi = s\bar{\psi}$, contradicting the nonsymmetry of ψ. In particular, ν cannot be a positive multiple of $\mathcal{H}_\psi$. □

Ahlfors regularity.

Corollary 9.5 (Ahlfors regularity)**.** *If Γ is Zariski dense, the following are equivalent:*

1. *The measure ν is Ahlfors s-regular for some $s > 0$;*
2. *The measure ν is Ahlfors 1-regular;*
3. *$\psi = \psi\circ\mathrm{i}$ and, for every $\mathsf{P} \in \mathcal{P}$, $\delta_\psi(\mathsf{P}) = \frac{1}{2}$ and $a_\psi(\mathsf{P}) = 0$.*

If ψ is symmetric, the equivalence also holds without assuming that Γ is Zariski dense. Here Ahlfors s-regularity means that there exists $C > 1$ such that

$$C^{-1}r^s \le \nu(B_\psi(\xi, r)) \le Cr^s \qquad (\xi \in \Lambda_\theta,\ 0 < r \le 1).$$

Proof. Suppose first that ν is Ahlfors s-regular. If Γ is Zariski dense, Ahlfors s-regularity gives $\nu \asymp \mathcal{H}^s_\psi$, so the proof of Theorem 9.1 forces $\psi = \psi \circ \mathrm{i}$. Thus, in either case covered by the statement, we may use the estimates for symmetric ψ. Choose a conical point $\xi \in \Lambda_\theta$. As in the proof of Lemma 9.3, there are points $x_i \in [o_Y, \xi]$ in the thick part of Y with $d(o_Y, x_i) \to \infty$. Set

$$r_i := e^{-\mathsf{d}_\psi(o_Y, x_i)}.$$

Then $r_i \to 0$, and the ball estimate in Remark 8.2, with $h_\psi(x_i) = 0$, gives the two-sided comparison $\nu(B_\psi(\xi, r_i)) \asymp r_i$, which implies $s = 1$.

Now fix $\mathsf{P} \in \mathcal{P}$ and its parabolic basepoint $p = \xi_{\mathsf{P}}$. For $x \in [o_Y, p]$ sufficiently far into its horoball, the comparison with closest orbit points in Theorem 7.1 and Lemma 4.11 give

$$h_\psi(x) \approx \mathsf{d}_\psi(\Gamma o_Y, x) = \mathsf{d}_\psi(x, \Gamma o_Y) \approx \mathsf{d}_\psi(x, o_Y) = \mathsf{d}_\psi(o_Y, x),$$

where the equalities use $\psi = \psi \circ \mathrm{i}$. Thus $h_\psi(x) = \mathsf{d}_\psi(o_Y, x) + O(1)$ along this ray. For every sufficiently small $r > 0$, choose $x \in [o_Y, p]$ with $\mathsf{d}_\psi(o_Y, x) = |\log r| + O(1)$ by Lemma 4.10(3). These points eventually lie in the horoball, and the preceding comparison gives $h_\psi(x) = |\log r| + O(1)$. Remark 8.2 therefore implies

$$\frac{\nu(B_\psi(p, r))}{r} \asymp r^{1-2\delta_\psi(\mathsf{P})}(1 + |\log r|)^{a_\psi(\mathsf{P})}, \tag{9.3}$$

with implied constants independent of r. If $\delta_\psi(\mathsf{P}) < 1/2$, the right-hand side tends to zero. If $\delta_\psi(\mathsf{P}) > 1/2$, or if $\delta_\psi(\mathsf{P}) = 1/2$ and $a_\psi(\mathsf{P}) > 0$, it tends to infinity. Since $a_\psi(\mathsf{P})$ is nonnegative, Ahlfors 1-regularity forces precisely the condition in (3). This shows (1) $\Rightarrow$ (2) $\Rightarrow$ (3).

Conversely, suppose that (3) holds. Every cusp correction factor in Remark 8.2 is then equal to 1. The same estimate holds in the thick part, so

$$\nu(B_\psi(\xi, r)) \asymp r \qquad (\xi \in \Lambda_\theta,\ 0 < r \le 1)$$

with uniform constants. This proves (2), and (2) implies (1). □

Cusped Hitchin representations and the middle simple root. Recall that a representation $\rho : \Gamma_0 \to \mathrm{PSL}(d, \mathbb{R})$ of a geometrically finite Fuchsian group Γ_0 containing parabolic elements is called *cusped Hitchin* if there exists a continuous ρ-equivariant positive map $\xi : \Lambda(\Gamma_0) \to \mathcal{F}(\mathbb{R}^d)$, where $\mathcal{F}(\mathbb{R}^d)$ is the full flag variety and positivity is understood in the sense of Fock–Goncharov; see [7, Section 1]. Equivalently, ρ is a type-preserving positive representation; see [7, Definition 9.1 and Theorem 9.2].

The peripheral condition in Theorem 9.1 can be checked explicitly for cusped Hitchin representations.

Corollary 9.6. *Let $\Gamma_0 < \mathrm{PSL}(2,\mathbb{R})$ be a torsion-free nonuniform lattice, let $m \geq 2$, and let $\rho : \Gamma_0 \to \mathrm{PSL}(2m,\mathbb{R})$ be a cusped Hitchin representation. Set $\Gamma := \rho(\Gamma_0)$, and $\theta = \Pi$. For*

$$\psi = \alpha_m, \qquad \alpha_m(\mathrm{diag}(t_1,\dots,t_{2m})) := t_m - t_{m+1},$$

the Patterson–Sullivan measure ν_ψ is Ahlfors 1-regular with respect to d_ψ. In particular, ν_ψ is a positive multiple of $\mathcal{H}_\psi$.

Proof. Since Γ_0 is a free group, ρ admits a lift to $\mathrm{SL}(2m,\mathbb{R})$, which we use in the Cartan calculations below. Moreover, Γ admits a relatively Π-Morse embedding f; see [30, Corollary 1.14] and [29, Proposition 11.3]. In particular, ψ is positive on $\mathcal{L}_f - \{0\}$. Since Γ_0 is a lattice, [8, Corollary 1.7] gives $\delta_\psi(\Gamma) = 1$. Moreover, ψ is invariant under i.

Each maximal peripheral subgroup of Γ is infinite cyclic. By [7, Theorem 1.4(2)], its generator u is a regular unipotent element, up to a central sign in a lift to $\mathrm{SL}(2m,\mathbb{R})$. If we put $H := \mathrm{diag}(2m-1, 2m-3, \dots, 1-2m)$, then

$$\mu(u^n) = (\log|n|)H + O(1) \qquad (|n| \to \infty).$$

Since $\alpha_m(H) = 2$, this gives

$$\psi(\mu(u^n)) = 2\log|n| + O(1).$$

The Poincaré series of $\langle u\rangle$ therefore has the same abscissa of convergence as $\sum_{n\neq 0}|n|^{-2s}$. Hence $\delta_\psi(\langle u\rangle) = \frac{1}{2}$ and $a_\psi(\langle u\rangle) = 0$, where the second assertion also follows from the virtually cyclic case of Theorem 5.1. Corollary 9.5 gives Ahlfors 1-regularity, and Theorem 9.1 gives the assertion about $\mathcal{H}_\psi$. □

Self-joinings. Consider a self-joining $\Gamma = (\rho_1 \times \cdots \times \rho_k)(\Gamma_0) < \prod_{i=1}^k \mathrm{SO}(n_i,1)^\circ$ as in the introduction: Γ_0 is geometrically finite, $\rho_1 = \mathrm{id}$, and the ρ_i are discrete faithful, geometrically finite, type-preserving representations. Here $\theta = \Pi$. We have $\mu(g) = (t_1,\dots,t_k)$ with $t_i = d_{\mathbb{H}^{n_i}}(o_i, g_i o_i)$. Write

$$\psi(t_1,\dots,t_k) = \sum_{i=1}^k b_i t_i, \qquad b_i > 0, \qquad \delta_\psi(\Gamma) = 1, \qquad B_\psi := \sum_{i=1}^k b_i.$$

In this case, i is trivial and hence ψ is symmetric. For a peripheral subgroup P, let $r(\mathsf{P})$ be the rank of a finite-index free abelian subgroup, and set $r_{\max} := \max_{\mathsf{P}\in\mathcal{P}} r(\mathsf{P})$, with $r_{\max} = 0$ if $\mathcal{P}$ is empty.

Corollary 9.7. *For every $\mathsf{P} \in \mathcal{P}$,*

$$\delta_\psi(\mathsf{P}) = \frac{r(\mathsf{P})}{2B_\psi}, \qquad a_\psi(\mathsf{P}) = 0.$$

Consequently, ν is a positive multiple of $\mathcal{H}_\psi$ if and only if $B_\psi \geq r_{\max}$. Moreover, ν is Ahlfors 1-regular with respect to d_ψ if and only if $r(\mathsf{P}) = B_\psi$ for every $\mathsf{P} \in \mathcal{P}$.

Proof. Write $r = r(\mathsf{P})$. Passing to a finite-index subgroup does not change the counting exponents. We may therefore assume that $\mathsf{P} = \{\gamma_n : n \in \mathbb{Z}^r\} \simeq \mathbb{Z}^r$ and that, in each factor, it acts by translations on an invariant r-dimensional affine subspace of a horosphere.

Choose a point q_i in this subspace. Normalizing the horosphere to have height 1, the hyperbolic distance formula gives

$$d_{\mathbb{H}^{n_i}}(q_i, (\gamma_n)_i q_i) = 2\log(1 + \|n\|) + O(1).$$

Replacing q_i by o_i changes this distance by at most $2d_{\mathbb{H}^{n_i}}(o_i, q_i)$. Hence

$$\psi(\mu(\gamma_n)) = 2B_\psi \log(1 + \|n\|) + O(1),$$

and counting lattice points gives

$$\#\{\gamma \in \mathsf{P} : \psi(\mu(\gamma)) \le T\} \asymp e^{rT/(2B_\psi)}.$$

This proves the formulas for $\delta_\psi(\mathsf{P})$ and $a_\psi(\mathsf{P})$. The remaining assertions follow from Theorem 9.1 and Corollary 9.5. □

Example 9.8 (Cusped surface self-joinings)**.** Let S be a once-punctured torus, and let $j_1, j_2 : \pi_1(S) \to \mathrm{PSL}_2(\mathbb{R})$ be two discrete faithful finite-area holonomies that are not conjugate in $\mathrm{PGL}_2(\mathbb{R})$. Set $\Gamma = (j_1 \times j_2)(\pi_1(S))$. For $0 < s < 1$, define

$$q_s(t_1, t_2) := s \cdot t_1 + (1 - s) \cdot t_2, \qquad \psi := \delta_{q_s}(\Gamma)\, q_s.$$

The coordinate critical exponents are both 1. Since j_1 and j_2 have different marked length spectra, [9, Corollary 1.9] gives $\delta_{q_s}(\Gamma) < 1$. By construction, $\delta_\psi(\Gamma) = 1$, every peripheral subgroup has rank 1, and $B_\psi = \delta_{q_s}(\Gamma) < 1$. Corollary 9.7 and Proposition 9.4 therefore give $\mathcal{H}_\psi(\Lambda_\theta) = 0$, although ν is locally doubling and locally reverse doubling by Theorem 8.3.

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School of Mathematics, Institute for Advanced Study, Princeton, NJ 08540 (Current)
Simons Laufer Mathematical Sciences Institute, Berkeley, CA 94720
Department of Mathematics, Yale University, New Haven, CT 06511
Email address: dongryul.kim97@gmail.com

Department of Mathematics, Yale University, New Haven, CT 06511
Email address: hee.oh@yale.edu